US012413183B2

(12) **United States Patent**
Clemente et al.

(10) **Patent No.: US 12,413,183 B2**
(45) **Date of Patent: Sep. 9, 2025**

(54) **ELECTRICAL CABLE PASSTHROUGH FOR PHOTOVOLTAIC SYSTEMS**

(71) Applicant: **GAF Energy LLC**, Parsippany, NJ (US)

(72) Inventors: **Peter Clemente**, Warren, NJ (US); **Gary Rossi**, San Francisco, CA (US); **Evan Michael Wray**, Cotati, CA (US)

(73) Assignee: **GAF Energy LLC**, Parsippany, NJ (US)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

(21) Appl. No.: **18/509,636**

(22) Filed: **Nov. 15, 2023**

(65) **Prior Publication Data**

US 2024/0162854 A1 May 16, 2024

Related U.S. Application Data

(60) Provisional application No. 63/383,844, filed on Nov. 15, 2022.

(51) **Int. Cl.**

H02G 3/22 (2006.01)
H02S 20/25 (2014.01)
H02S 40/30 (2014.01)

(52) **U.S. Cl.**

CPC ***H02S 40/30*** (2014.12); ***H02G 3/22*** (2013.01); ***H02S 20/25*** (2014.12)

(58) **Field of Classification Search**

CPC .. H02G 3/06–088; H02G 3/22; H05K 5/0247; H02S 20/25; H02S 40/30

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

1,981,467	A	11/1934	Radtke
3,156,497	A	11/1964	Lessard
3,581,779	A	6/1971	Gilbert, Jr.
4,258,948	A	3/1981	Hoffmann
4,349,220	A	9/1982	Carroll et al.
4,499,702	A	2/1985	Turner
4,636,577	A	1/1987	Peterpaul

(Continued)

FOREIGN PATENT DOCUMENTS

CA	2829440	A	5/2019
CH	700095	A2	6/2010

(Continued)

OTHER PUBLICATIONS

Sunflare, Procducts: “Sunflare Develops Prototype For New Residential Solar Shingles”; 2019 <<sunflaresolar.com/news/sunflare-develops-prototype-for-new-residential-solar-shingles>> retrieved Feb. 2, 2021.

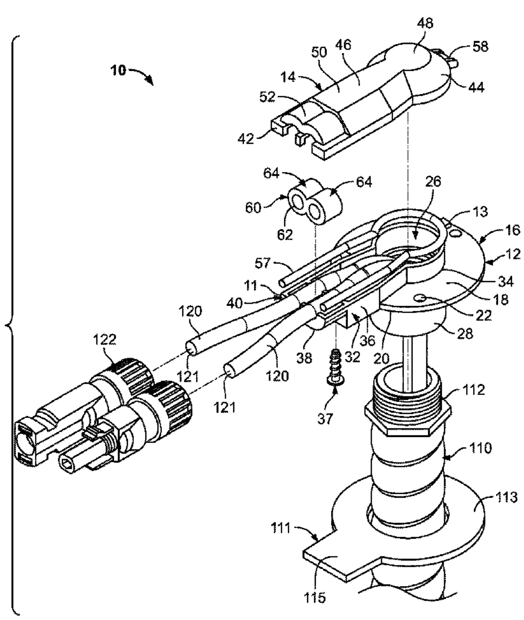
(Continued)

Primary Examiner — Roshn K Varghese
(74) *Attorney, Agent, or Firm* — GREENBERG TRAURIG, LLP

(57) **ABSTRACT**

A system comprises a passthrough installed on a roof deck. The passthrough includes a base with an aperture, and a lid attached to the base. The aperture of the base is configured to align with an aperture formed within the roof deck. The system includes at least one cable and a grommet installed on the at least one cable. The lid and the base are configured to compress the grommet around the at least one cable.

15 Claims, 9 Drawing Sheets



(56) References Cited

U.S. PATENT DOCUMENTS

5,167,579	A	12/1992	Rotter
5,437,735	A	8/1995	Younan et al.
5,590,495	A	1/1997	Bressler et al.
5,642,596	A	7/1997	Waddington
6,008,450	A	12/1999	Ohtsuka et al.
6,033,270	A	3/2000	Stuart
6,046,399	A	4/2000	Kapner
6,201,180	B1	3/2001	Meyer et al.
6,220,329	B1	4/2001	King et al.
6,308,482	B1	10/2001	Strait
6,320,114	B1	11/2001	Kuechler
6,320,115	B1	11/2001	Kataoka et al.
6,336,304	B1	1/2002	Mimura et al.
6,341,454	B1	1/2002	Koleoglou
6,407,329	B1	6/2002	Iino et al.
6,576,830	B2	6/2003	Nagao et al.
6,793,940	B2	9/2004	Tournilhac et al.
6,928,781	B2	8/2005	Desbois et al.
6,972,367	B2	12/2005	Federspiel et al.
7,138,578	B2	11/2006	Komamine
7,155,870	B2	1/2007	Almy
7,178,295	B2	2/2007	Dinwoodie
7,487,771	B1	2/2009	Eifert et al.
7,587,864	B2	9/2009	McCaskill et al.
7,678,990	B2	3/2010	McCaskill et al.
7,678,991	B2	3/2010	McCaskill et al.
7,748,191	B2	7/2010	Podirsky
7,819,114	B2	10/2010	Augenbraun et al.
7,824,191	B1	11/2010	Podirsky
7,832,176	B2	11/2010	McCaskill et al.
8,118,109	B1	2/2012	Hacker
8,168,880	B2	5/2012	Jacobs et al.
8,173,889	B2	5/2012	Kalkanoglu et al.
8,210,570	B1	7/2012	Railkar et al.
8,276,329	B2	10/2012	Lenox
8,312,693	B2	11/2012	Cappelli
8,319,093	B2	11/2012	Kalkanoglu et al.
8,333,040	B2	12/2012	Shiao et al.
8,371,076	B2	2/2013	Jones et al.
8,375,653	B2	2/2013	Shiao et al.
8,404,967	B2	3/2013	Kalkanoglu et al.
8,410,349	B2	4/2013	Kalkanoglu et al.
8,418,415	B2	4/2013	Shiao et al.
8,438,796	B2	5/2013	Shiao et al.
8,468,754	B2	6/2013	Railkar et al.
8,468,757	B2	6/2013	Krause et al.
8,505,249	B2	8/2013	Geary
8,512,866	B2	8/2013	Taylor
8,513,517	B2	8/2013	Kalkanoglu et al.
8,586,856	B2	11/2013	Kalkanoglu et al.
8,601,754	B2	12/2013	Jenkins et al.
8,629,578	B2	1/2014	Kurs et al.
8,646,228	B2	2/2014	Jenkins
8,656,657	B2	2/2014	Livsey et al.
8,671,630	B2	3/2014	Lena et al.
8,677,702	B2	3/2014	Jenkins
8,695,289	B2	4/2014	Koch et al.
8,713,858	B1	5/2014	Xie
8,713,860	B2	5/2014	Railkar et al.
8,733,038	B2	5/2014	Kalkanoglu et al.
8,776,455	B2	7/2014	Azoulay
8,789,321	B2	7/2014	Ishida
8,793,941	B2	8/2014	Bosler et al.
8,826,607	B2	9/2014	Shiao et al.
8,835,751	B2	9/2014	Kalkanoglu et al.
8,863,451	B2	10/2014	Jenkins et al.
8,898,970	B2	12/2014	Jenkins et al.
8,925,262	B2	1/2015	Railkar et al.
8,943,766	B2	2/2015	Gombarick et al.
8,946,544	B2	2/2015	Jabos et al.
8,950,128	B2	2/2015	Kalkanoglu et al.
8,959,848	B2	2/2015	Jenkins et al.
8,966,838	B2	3/2015	Jenkins
8,966,850	B2	3/2015	Jenkins et al.
8,994,224	B2	3/2015	Mehta et al.
9,032,672	B2	5/2015	Livsey et al.
9,153,950	B2	10/2015	Yamanaka et al.
9,166,087	B2	10/2015	Chihlas et al.
9,169,646	B2	10/2015	Rodrigues et al.
9,170,034	B2	10/2015	Bosler et al.
9,178,465	B2	11/2015	Shiao et al.
9,202,955	B2	12/2015	Livsey et al.
9,212,832	B2	12/2015	Jenkins
9,217,584	B2	12/2015	Kalkanoglu et al.
9,270,221	B2	2/2016	Zhao
9,273,885	B2	3/2016	Rordigues et al.
9,276,141	B2	3/2016	Kalkanoglu et al.
9,331,224	B2	5/2016	Koch et al.
9,356,174	B2	5/2016	Duarte et al.
9,359,014	B1	6/2016	Yang et al.
9,412,890	B1	8/2016	Meyers
9,528,270	B2	12/2016	Jenkins et al.
9,605,432	B1	3/2017	Robbins
9,711,672	B2	7/2017	Wang
9,755,573	B2	9/2017	Livsey et al.
9,786,802	B2	10/2017	Shiao et al.
9,831,818	B2	11/2017	West
9,912,284	B2	3/2018	Svec
9,923,515	B2	3/2018	Rodrigues et al.
9,938,729	B2	4/2018	Coon
9,991,412	B2	6/2018	Gonzalez et al.
9,998,067	B2	6/2018	Kalkanoglu et al.
10,027,273	B2	7/2018	West et al.
10,115,850	B2	10/2018	Rodrigues et al.
10,128,660	B1	11/2018	Apte et al.
10,156,075	B1	12/2018	McDonough
10,187,005	B2	1/2019	Rodrigues et al.
10,256,765	B2	4/2019	Rodrigues et al.
10,284,136	B1	5/2019	Mayfield et al.
10,454,408	B2	10/2019	Livsey et al.
10,530,292	B1	1/2020	Cropper et al.
10,560,048	B2	2/2020	Fisher et al.
10,563,406	B2	2/2020	Kalkanoglu et al.
D879,031	S	3/2020	Lance et al.
10,579,028	B1	3/2020	Jacob
10,784,813	B2	9/2020	Kalkanoglu et al.
D904,289	S	12/2020	Lance et al.
11,012,026	B2	5/2021	Kalkanoglu et al.
11,177,639	B1	11/2021	Nguyen et al.
11,217,715	B2	1/2022	Sharenko
11,251,744	B1	2/2022	Bunea
11,258,399	B2	2/2022	Kalkanoglu et al.
11,283,394	B2	3/2022	Perkins et al.
11,309,828	B2	4/2022	Sirski et al.
11,394,344	B2	7/2022	Perkins et al.
11,424,379	B2	8/2022	Sharenko et al.
11,431,280	B2	8/2022	Liu et al.
11,431,281	B2	8/2022	Perkins et al.
11,444,569	B2	9/2022	Clemente et al.
11,454,027	B2	9/2022	Kuiper et al.
11,459,757	B2	10/2022	Nguyen et al.
11,486,144	B2	11/2022	Bunea et al.
11,489,482	B2	11/2022	Peterson et al.
11,496,088	B2	11/2022	Sirski et al.
11,508,861	B1	11/2022	Perkins et al.
11,512,480	B1	11/2022	Achor et al.
11,527,665	B2	12/2022	Boitnott
11,545,927	B2	1/2023	Abra et al.
11,545,928	B2	1/2023	Perkins et al.
11,658,470	B2	5/2023	Nguyen et al.
11,661,745	B2	5/2023	Bunea et al.
11,689,149	B2	6/2023	Clemente et al.
11,705,531	B2	7/2023	Sharenko et al.
11,728,759	B2	8/2023	Nguyen et al.
11,732,490	B2	8/2023	Achor et al.
11,811,361	B1	11/2023	Farhangi et al.
11,824,486	B2	11/2023	Nguyen et al.
11,824,487	B2	11/2023	Nguyen et al.
11,843,067	B2	12/2023	Nguyen et al.
2002/0053360	A1	5/2002	Kinoshita et al.
2002/0129849	A1	9/2002	Heckeroth
2003/0101662	A1	6/2003	Ullman
2003/0132265	A1	7/2003	Villela et al.
2003/0217768	A1	11/2003	Guha

(56) **References Cited**

U.S. PATENT DOCUMENTS

2004/0000334 A1	1/2004	Ressler	
2005/0030187 A1	2/2005	Peress et al.	
2005/0115603 A1	6/2005	Yoshida et al.	
2005/0144670 A1	6/2005	Fujiyama et al.	
2005/0178428 A1	8/2005	Laaly et al.	
2005/0193673 A1	9/2005	Rodrigues et al.	
2006/0042683 A1	3/2006	Gangemi	
2006/0046084 A1	3/2006	Yang et al.	
2007/0007038 A1*	1/2007	Cox	H02G 15/013 174/650
2007/0074757 A1	4/2007	Mellott et al.	
2007/0181174 A1	8/2007	Ressler	
2007/0193618 A1	8/2007	Bressler et al.	
2007/0216110 A1*	9/2007	Stuckmann	H02G 3/22 277/607
2007/0249194 A1	10/2007	Liao	
2007/0295385 A1	12/2007	Sheats et al.	
2008/0006323 A1	1/2008	Kalkanoglu et al.	
2008/0035140 A1	2/2008	Placer et al.	
2008/0078440 A1	4/2008	Lim et al.	
2008/0185748 A1	8/2008	Kalkanoglu	
2008/0271774 A1	11/2008	Kalkanoglu et al.	
2008/0302030 A1	12/2008	Stancel et al.	
2008/0315061 A1	12/2008	Fath	
2009/0000222 A1	1/2009	Kalkanoglu et al.	
2009/0014057 A1	1/2009	Croft et al.	
2009/0014058 A1	1/2009	Croft et al.	
2009/0019795 A1	1/2009	Szacsvay et al.	
2009/0044850 A1	2/2009	Kimberley	
2009/0114261 A1	5/2009	Stancel et al.	
2009/0133340 A1	5/2009	Shiao et al.	
2009/0159118 A1	6/2009	Kalkanoglu et al.	
2009/0178350 A1	7/2009	Kalkanoglu et al.	
2009/0229652 A1	9/2009	Mapel et al.	
2009/0275247 A1	11/2009	Richter et al.	
2010/0019580 A1	1/2010	Croft et al.	
2010/0095618 A1	4/2010	Edison et al.	
2010/0101634 A1	4/2010	Frank et al.	
2010/0116325 A1	5/2010	Nikoonahad	
2010/0131108 A1	5/2010	Meyer	
2010/0139184 A1	6/2010	Williams et al.	
2010/0146878 A1	6/2010	Koch et al.	
2010/0159221 A1	6/2010	Kourtakis et al.	
2010/0170169 A1	7/2010	Railkar et al.	
2010/0186798 A1	7/2010	Tormen et al.	
2010/0242381 A1	9/2010	Jenkins	
2010/0313499 A1	12/2010	Gangemi	
2010/0325976 A1	12/2010	DeGenfelder et al.	
2010/0326488 A1	12/2010	Aue et al.	
2010/0326501 A1	12/2010	Zhao et al.	
2011/0030761 A1	2/2011	Kalkanoglu et al.	
2011/0036386 A1	2/2011	Browder	
2011/0036389 A1	2/2011	Hardikar et al.	
2011/0048507 A1	3/2011	Livsey et al.	
2011/0058337 A1	3/2011	Han et al.	
2011/0061326 A1	3/2011	Jenkins	
2011/0100436 A1	5/2011	Cleereman et al.	
2011/0104488 A1	5/2011	Muessig et al.	
2011/0132427 A1	6/2011	Kalkanoglu et al.	
2011/0168238 A1	7/2011	Metin et al.	
2011/0239555 A1	10/2011	Cook et al.	
2011/0302859 A1	12/2011	Crasnianski	
2011/0314753 A1	12/2011	Farmer et al.	
2012/0034799 A1	2/2012	Hunt	
2012/0060434 A1	3/2012	Jacobs	
2012/0060902 A1	3/2012	Drake	
2012/0085392 A1	4/2012	Albert et al.	
2012/0137600 A1	6/2012	Jenkins	
2012/0176077 A1	7/2012	Oh et al.	
2012/0212065 A1	8/2012	Cheng et al.	
2012/0233940 A1	9/2012	Perkins et al.	
2012/0240490 A1	9/2012	Gangemi	
2012/0260977 A1	10/2012	Stancel	
2012/0266942 A1	10/2012	Komatsu et al.	
2012/0279150 A1	11/2012	Pisikak et al.	
2012/0282437 A1	11/2012	Clark et al.	
2012/0291848 A1	11/2012	Sherman et al.	
2013/0008499 A1	1/2013	Verger et al.	
2013/0014455 A1	1/2013	Grieco	
2013/0118558 A1	5/2013	Sherman	
2013/0193769 A1	8/2013	Mehta et al.	
2013/0247988 A1	9/2013	Reese et al.	
2013/0284267 A1	10/2013	Plug et al.	
2013/0306137 A1	11/2013	Ko	
2014/0090697 A1	4/2014	Rodrigues et al.	
2014/0150843 A1	6/2014	Pearce et al.	
2014/0173997 A1	6/2014	Jenkins	
2014/0179220 A1	6/2014	Railkar et al.	
2014/0182222 A1	7/2014	Kalkanoglu et al.	
2014/0208675 A1	7/2014	Beerer et al.	
2014/0254776 A1	9/2014	O'Connor et al.	
2014/0266289 A1	9/2014	Della Sera et al.	
2014/0284098 A1	9/2014	Yamanaka et al.	
2014/0311556 A1	10/2014	Feng et al.	
2014/0352760 A1	12/2014	Haynes et al.	
2014/0366464 A1	12/2014	Rodrigues et al.	
2015/0089895 A1	4/2015	Leitch	
2015/0162459 A1	6/2015	Lu et al.	
2015/0340516 A1	11/2015	Kim et al.	
2015/0349173 A1	12/2015	Morad et al.	
2016/0105144 A1	4/2016	Haynes et al.	
2016/0142008 A1	5/2016	Lopez et al.	
2016/0254776 A1	9/2016	Rodrigues et al.	
2016/0276508 A1	9/2016	Huang et al.	
2016/0359451 A1	12/2016	Mao et al.	
2017/0159292 A1	6/2017	Chihlas et al.	
2017/0179319 A1	6/2017	Yamashita et al.	
2017/0179726 A1	6/2017	Garrity et al.	
2017/0237390 A1	8/2017	Hudson et al.	
2017/0331415 A1	11/2017	Koppi et al.	
2018/0094438 A1	4/2018	Wu et al.	
2018/0097472 A1	4/2018	Anderson et al.	
2018/0115275 A1	4/2018	Flanigan et al.	
2018/0254738 A1	9/2018	Yang et al.	
2018/0294765 A1	10/2018	Friedrich et al.	
2018/0351502 A1	12/2018	Almy et al.	
2018/0367089 A1	12/2018	Stutterheim et al.	
2019/0030867 A1	1/2019	Sun et al.	
2019/0081436 A1	3/2019	Onodi et al.	
2019/0123679 A1	4/2019	Rodrigues et al.	
2019/0253022 A1	8/2019	Hardar et al.	
2019/0305717 A1	10/2019	Allen et al.	
2020/0109320 A1	4/2020	Jiang	
2020/0144958 A1	5/2020	Rodrigues et al.	
2020/0220819 A1	7/2020	Vu et al.	
2020/0224419 A1	7/2020	Boss et al.	
2020/0343397 A1	10/2020	Hem-Jensen	
2021/0083619 A1	3/2021	Hegedus	
2021/0115223 A1	4/2021	Bonekamp et al.	
2021/0159353 A1	5/2021	Li et al.	
2021/0301536 A1	9/2021	Baggs et al.	
2021/0343886 A1	11/2021	Sharenko et al.	
2022/0149213 A1	5/2022	Mensink et al.	

FOREIGN PATENT DOCUMENTS

CN	202797032	U	3/2013
CN	217150978	U	8/2022
DE	1958248	A1	11/1971
EP	1039361	A1	9/2000
EP	1837162	A1	9/2007
EP	1774372	A1	7/2011
EP	2446481	A2	5/2012
EP	2784241	A1	10/2014
EP	3772175	A1	2/2021
JP	H02243866	A	9/1990
JP	10046767	A	2/1998
JP	2001-248276	A	9/2001
JP	2002-106151	A	4/2002
JP	2001-098703	A	10/2002
JP	2008-266977	A	11/2008
JP	2014-233153	A	12/2014
JP	2015-040457	A	3/2015
JP	2017-027735	A	2/2017

(56) **References Cited**

FOREIGN PATENT DOCUMENTS

JP	2018053707	A	4/2018
KR	20090084060	A	8/2009
KR	10-1348283	B1	1/2014
KR	10-2019-0000367	A	1/2019
KR	10-2253483	B1	5/2021
NL	2026856	B1	6/2022
WO	2010/151777	A2	12/2010
WO	2011/049944	A1	4/2011
WO	2015/133632	A1	9/2015
WO	2018/000589	A1	1/2018
WO	2019/201416	A1	10/2019
WO	2020-159358	A1	8/2020
WO	2021-247098	A1	12/2021

OTHER PUBLICATIONS

RGS Energy, 3.5kW Powerhouse 3.0 system installed in an afternoon; Jun. 7, 2019 <<facebook.com/RGSEnergy/>> retrieved Feb. 2, 2021.
Tesla, Solar Roof <<tesla.com/solarroof>> retrieved Feb. 2, 2021.
"Types of Roofing Underlayment", Owens Corning Roofing; <<https://www.owenscorning.com/en-us/roofing/tools/how-roofing-underlayment-helps-protect-your-home>> retrieved Nov. 1, 2021.

* cited by examiner

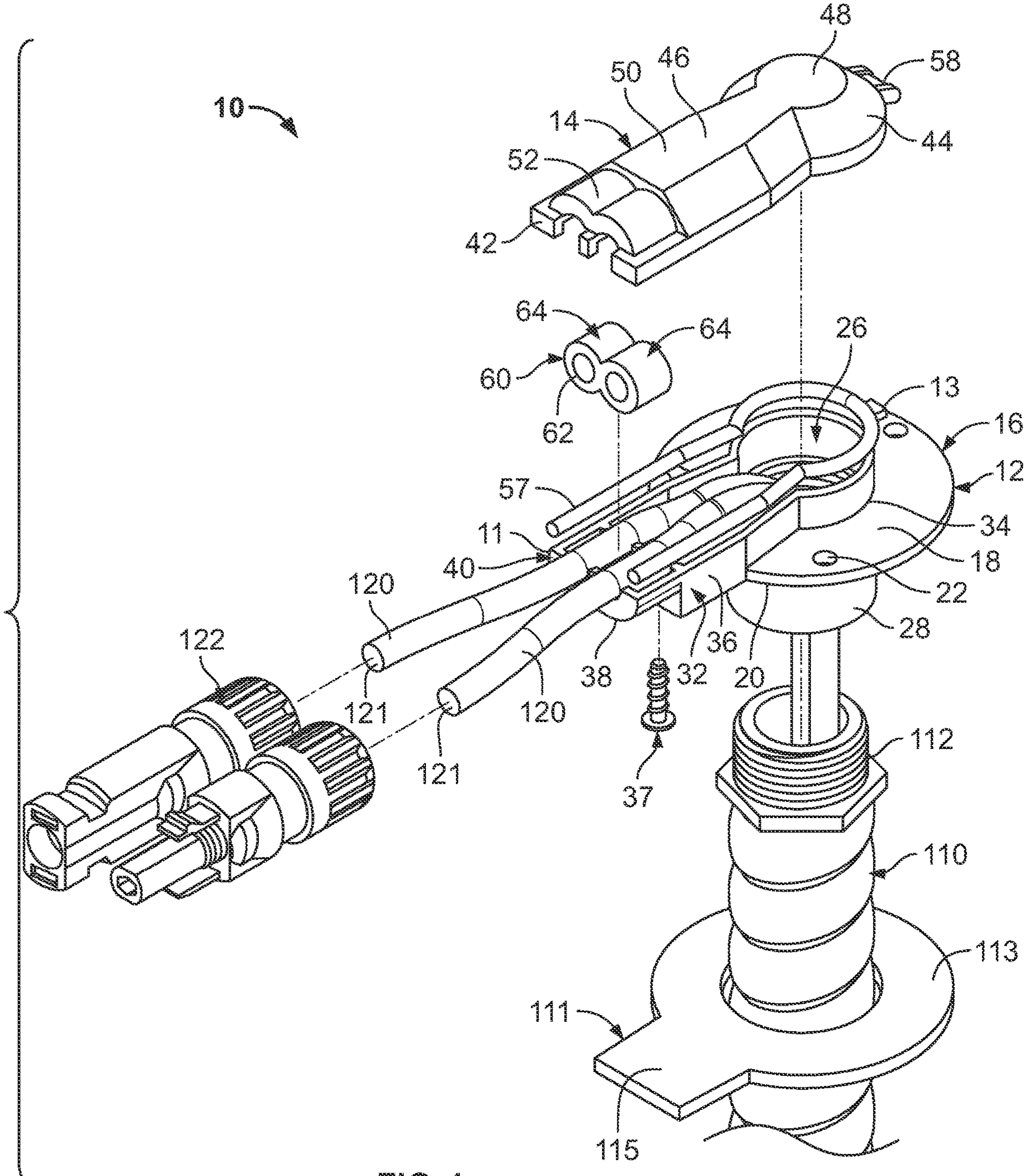

10
14
46
48
58
50
44
52
42
64
64
60
62
26
13
16
12
57
34
11
18
40
22
120
36
28
32
20
38
122
121
120
121
37
112
110
113
111
115

FIG. 1

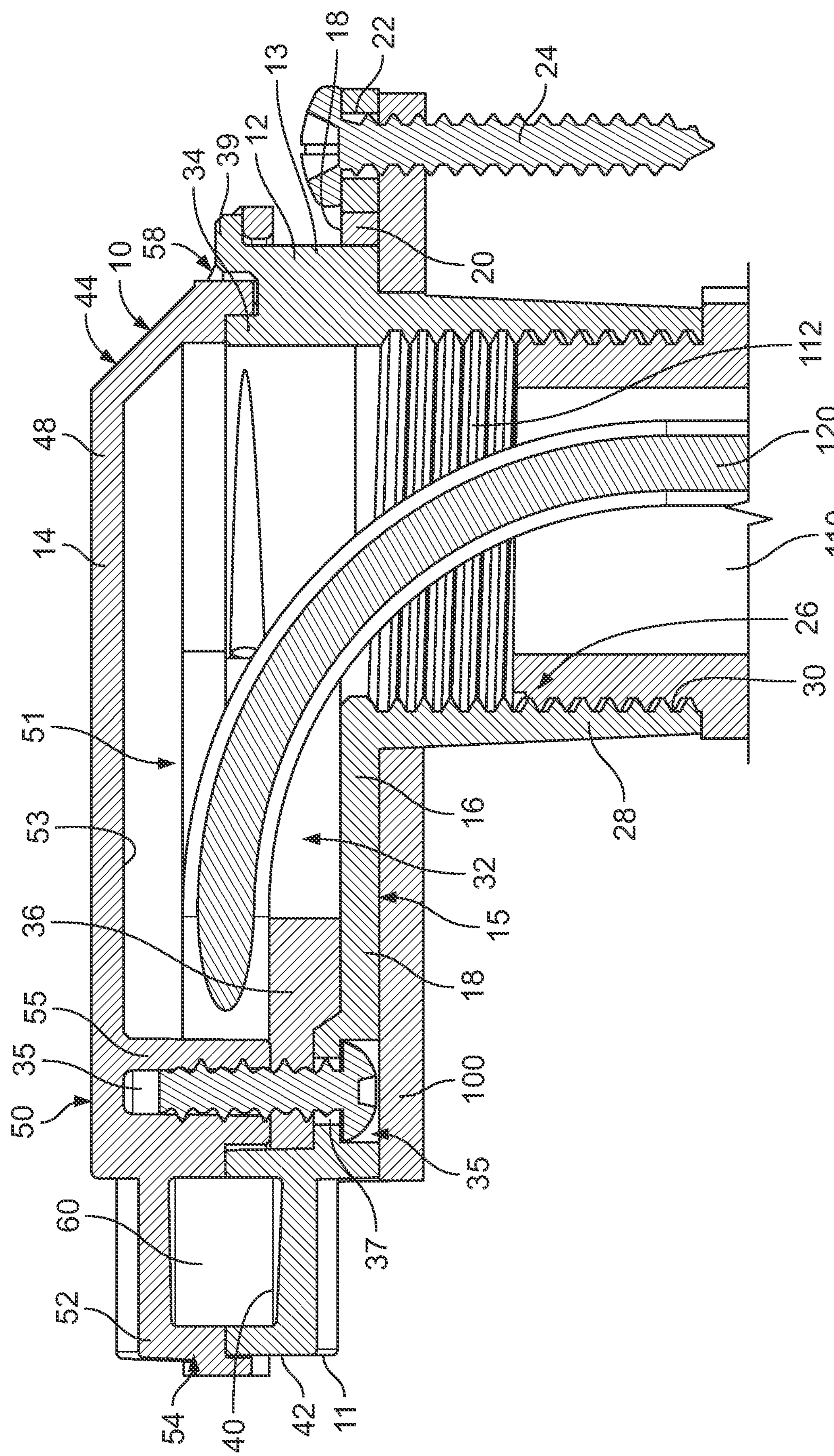

18
22
24
13
12
39
34
58
10
44
20
112
48
120
14
110
26
30
51
28
16
53
32
15
36
18
55
100
35
50
35
60
37
52
54
40
42
11

FIG. 2

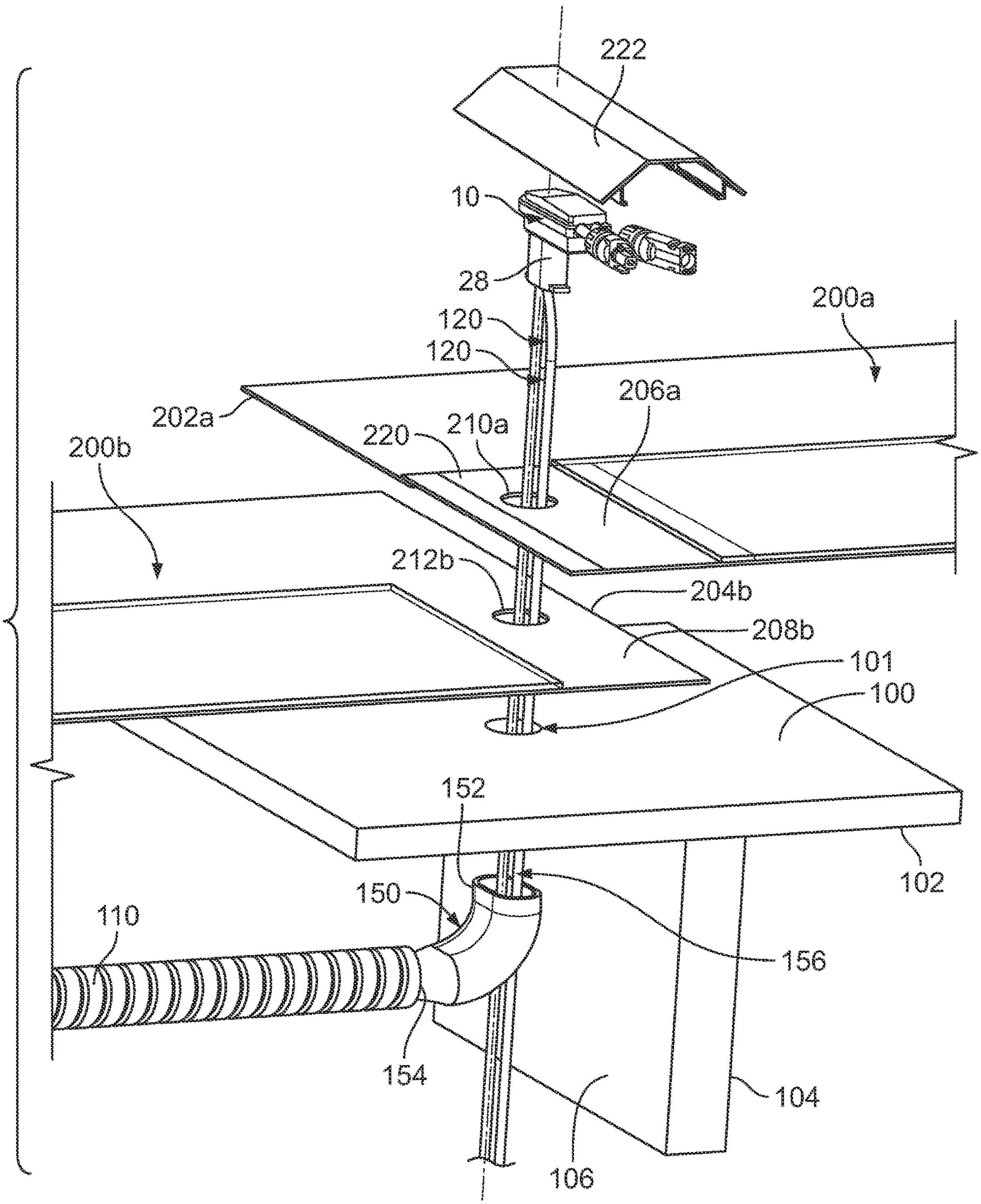

222
10
28
120
120
200a
206a
202a
220
210a
200b
212b
204b
208b
101
100
152
102
150
110
156
154
104
106

FIG. 3A

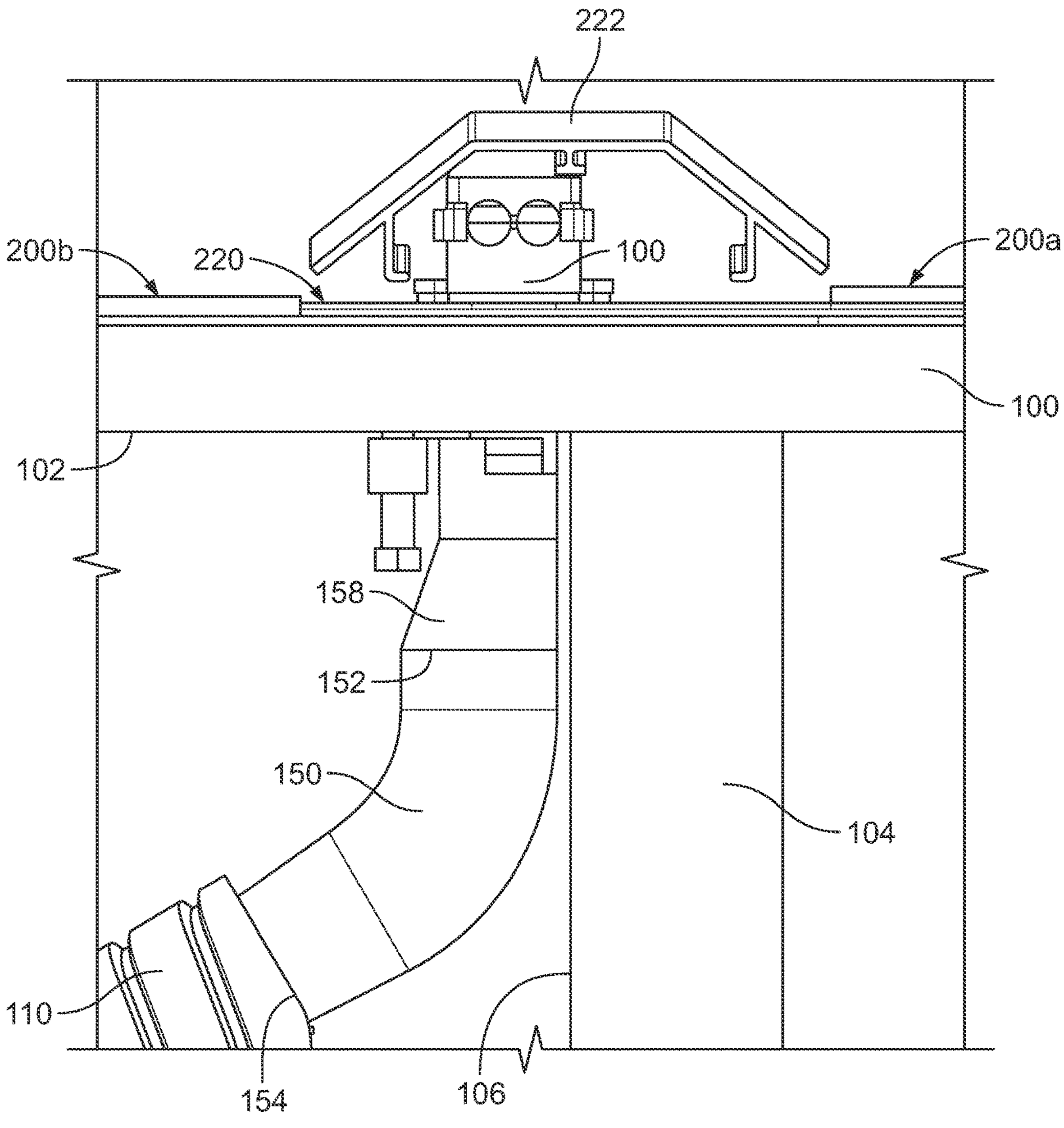


FIG. 3B

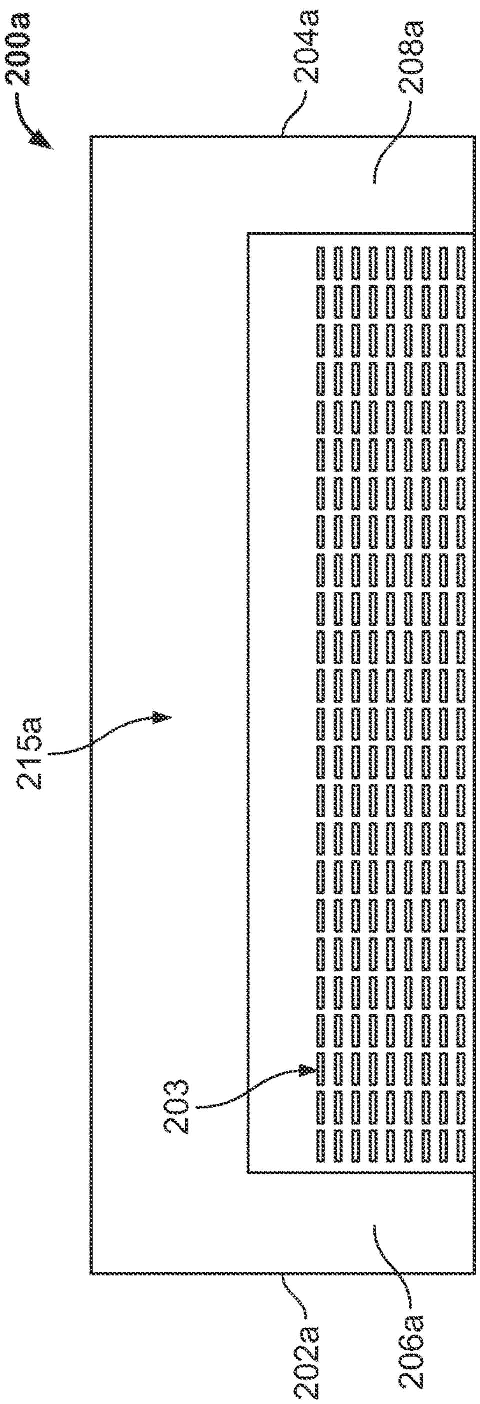

200a
204a
208a
215a
203
202a
206a

FIG. 3C

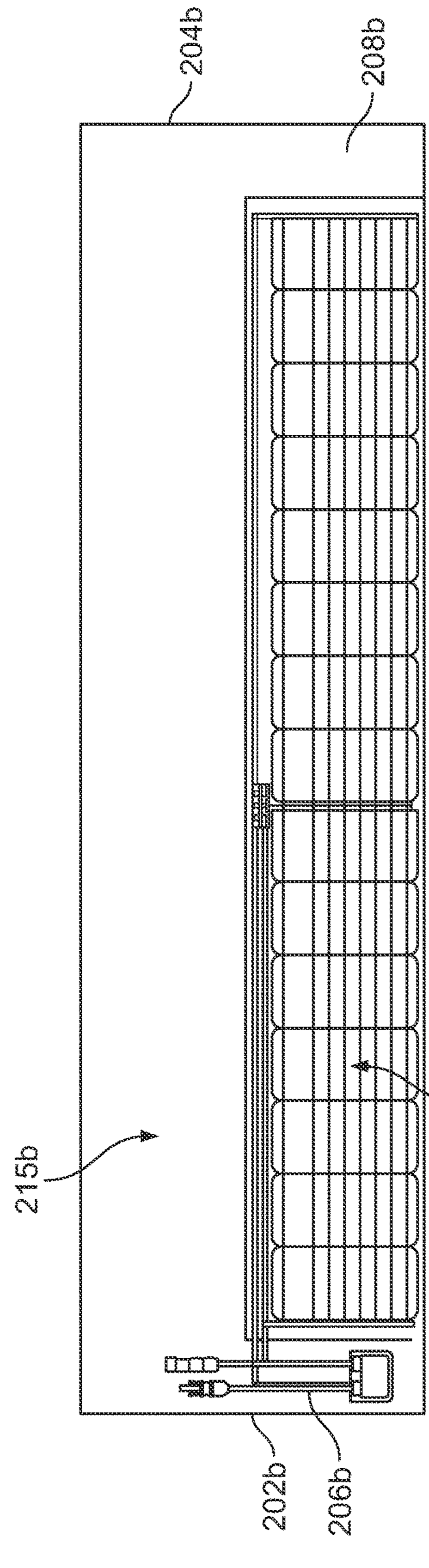

204b
208b
215b
201
202b
206b

FIG. 3D

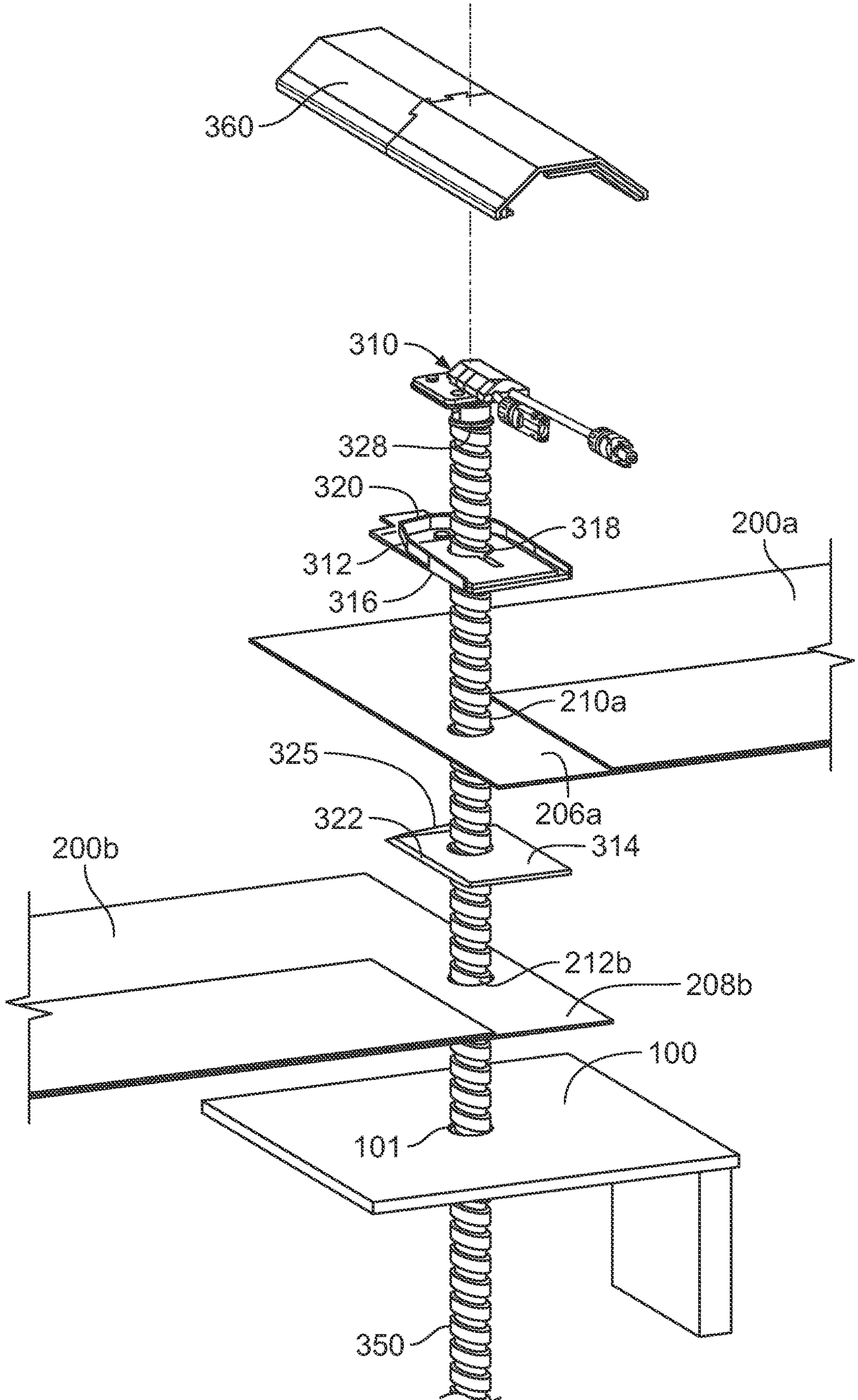

360
310
328
320
318
200a
312
316
210a
325
206a
322
314
200b
212b
208b
100
101
350

FIG. 4

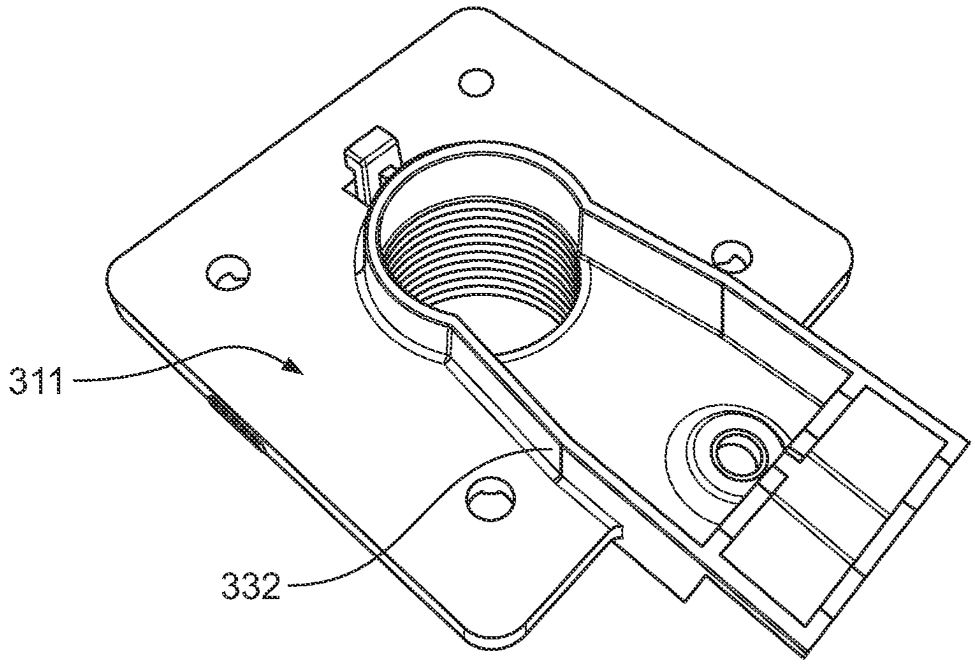


FIG. 4A

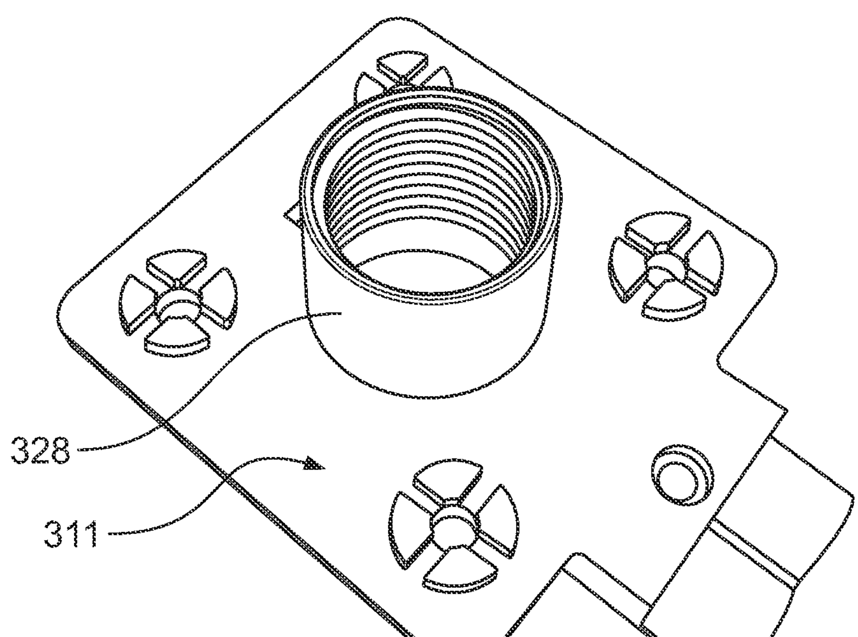


FIG. 4B

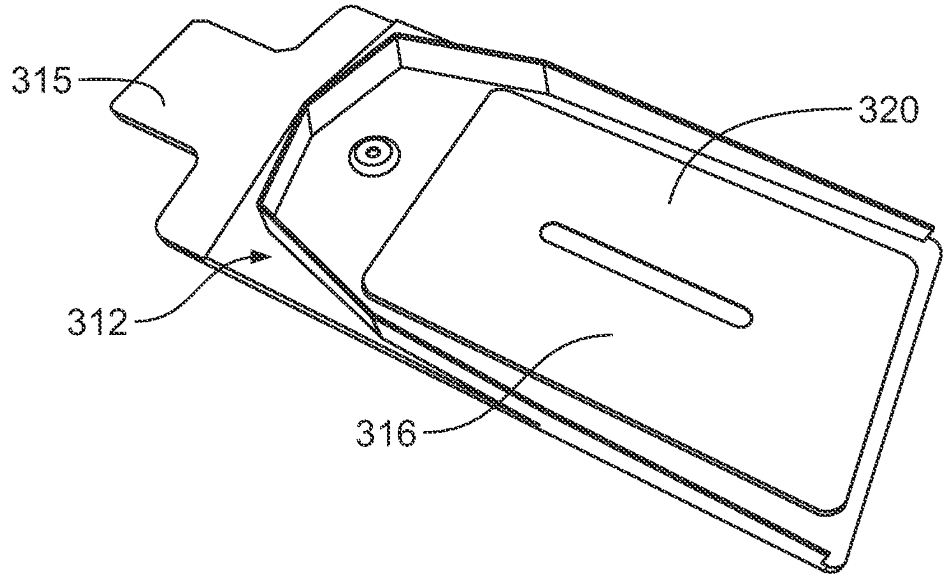


FIG. 4C

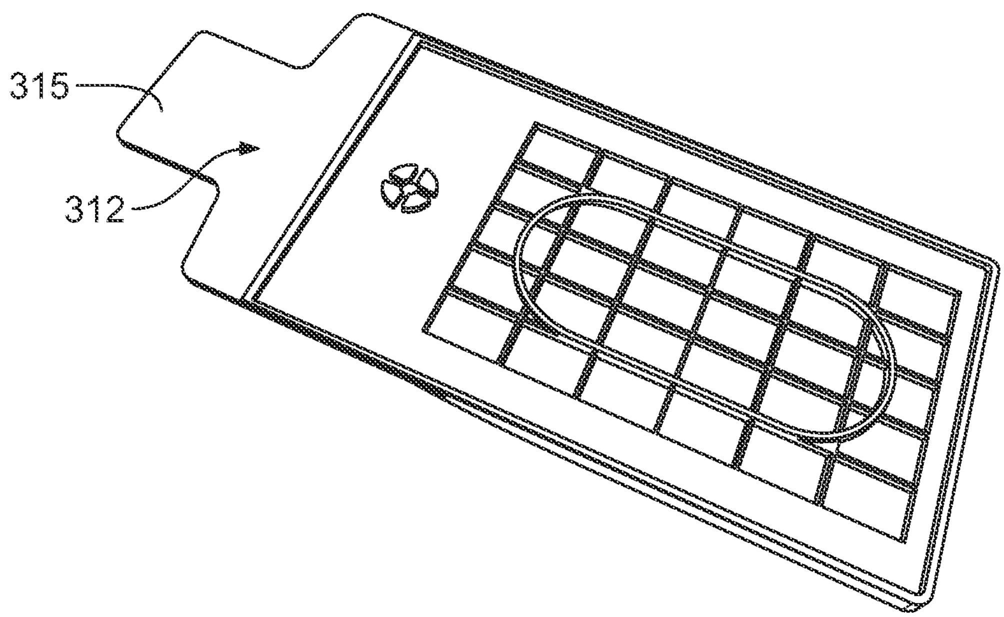


FIG. 4D

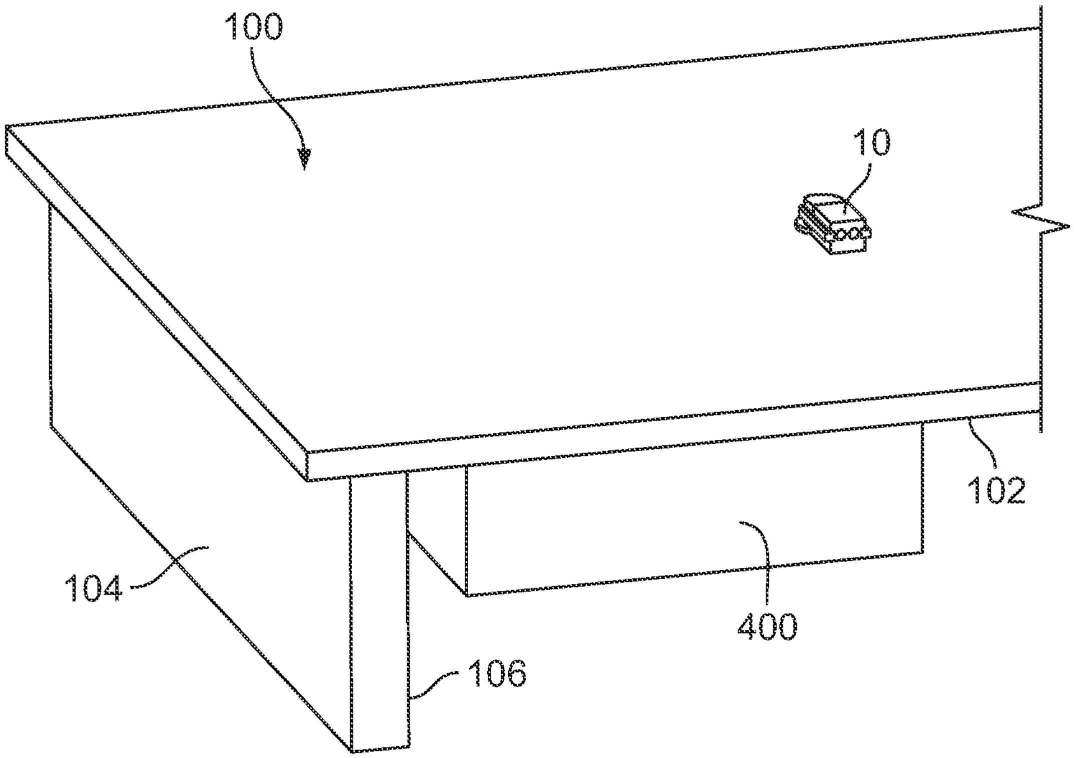


FIG. 5

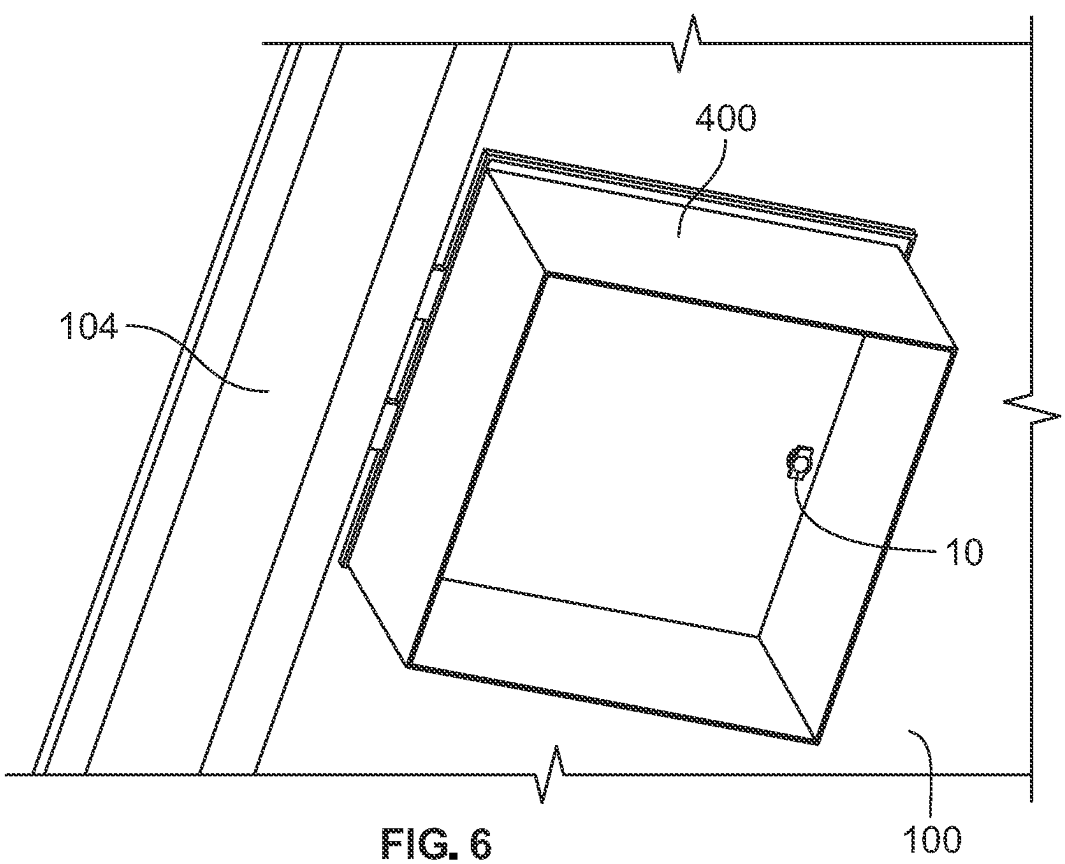


FIG. 6

ELECTRICAL CABLE PASSTHROUGH FOR PHOTOVOLTAIC SYSTEMS

CROSS-REFERENCE TO RELATED APPLICATION

This application is a Section 111(a) application relating to and claiming the benefit of commonly-owned, U.S. Provisional Patent Application Ser. No. 63/383,844, filed Nov. 15, 2022, entitled "ELECTRICAL CABLE PASSTHROUGH FOR PHOTOVOLTAIC SYSTEMS," the contents of which is incorporated herein by reference in its entirety.

FIELD OF THE INVENTION

The present invention relates to passthrough devices and, more particularly, passthrough devices for electrical cables for photovoltaic systems.

BACKGROUND OF THE INVENTION

Photovoltaic systems having solar modules are commonly installed on roofing of structures.

SUMMARY

In some embodiments, a system includes a roof deck; a passthrough installed on the roof deck, wherein the passthrough includes a base having a first end and a second end opposite the first end, a first surface extending from the first end to the second end, a second surface opposite the first surface, and an aperture extending from the first surface to the second surface, a lid attached to the base, wherein the lid includes a first end substantially aligned with the first end of the base, and a sidewall between the base and the lid, wherein the first surface is juxtaposed with the roof deck, and wherein the aperture of the base of the passthrough is configured to align with an aperture formed within the roof deck; at least one cable, wherein the at least one cable includes a first end and a second end opposite the first end of the at least one cable; and a grommet installed on the at least one cable proximate to the first end of the at least one cable, wherein the lid and the base are configured to compress the grommet around the at least one cable when the lid is attached to the base, wherein the aperture of the base of the passthrough is sized and shaped to receive the at least one cable, and wherein the first end of the at least one cable extends outwardly from the first end of base and the first end of the lid of the passthrough.

In some embodiments, the lid is removably attached to the base. In some embodiments, the system includes a photovoltaic module installed on the roof deck, wherein the passthrough is installed proximate to the photovoltaic module. In some embodiments, the sidewall includes at least one aperture, wherein the at least one aperture of the sidewall is sized and shaped to receive a corresponding one of the at least one cable. In some embodiments, the grommet includes at least one opening sized and shaped to receive a corresponding one of the at least one cable. In some embodiments, the sidewall includes a first end, wherein the first end of the sidewall is proximate to the first end of the base, and wherein the at least one aperture of the sidewall is located at the first end of the sidewall. In some embodiments, the base includes at least one first cavity, wherein the lid includes at least one second cavity, and wherein the at least one first cavity and the at least one second cavity are sized and shaped to form a corresponding one of the at least one aperture. In some embodiments, the grommet is located between the at least one first cavity and the at least one second cavity.

In some embodiments, the grommet is configured to be compressed between the at least one first cavity and the at least one second cavity. In some embodiments, the at least one first cavity includes a plurality of first cavities, wherein the at least one second cavity includes a plurality of second cavities, and wherein the at least one aperture of the sidewall includes a plurality of apertures, wherein the at least one cable includes a plurality of cables. In some embodiments, the at least one opening of the grommet includes a plurality of openings, and wherein each of the plurality of openings is sized and shaped to receive a corresponding one of the plurality of cables. In some embodiments, the grommet includes a plurality of sections, wherein each of the plurality of openings is located in a corresponding one of the plurality of sections, and wherein each of the plurality of sections of the grommet is located between a corresponding one of the plurality of first cavities and a corresponding one of the plurality of second cavities. In some embodiments, the at least one cable includes an electrical connector, wherein the at least one aperture of the sidewall is sized and shaped to receive the electrical connector of the at least one cable. In some embodiments, the base of the passthrough includes a tubular member extending from the first surface, wherein the tubular member includes the aperture of the base, and wherein the aperture of the roof deck is sized and shaped to receive the tubular member.

In some embodiments, the system further includes a receiver attached to the tubular member. In some embodiments, the system further includes an electrical box installed underneath the roof deck, and wherein the electrical box directly receives the tubular member.

In some embodiments, a cable passthrough includes a base having a first end and a second end opposite the first end, a first surface extending from the first end to the second end, a second surface opposite the first surface, and an aperture extending from the first surface to the second surface; a lid attached to the base, wherein the lid includes a first end substantially aligned with the first end of the base; and a sidewall between the base and the lid, wherein the cable passthrough is configured to be installed on a roof deck such that the first surface is juxtaposed with the roof deck, and wherein the aperture of the base is configured to align with an aperture formed within the roof deck, wherein the cable passthrough is configured to receive at least one cable, wherein the aperture of the base is sized and shaped to receive the at least one cable, wherein the at least one cable is capable of extending outwardly from the first end of the base and the first end of the lid of the cable passthrough, and wherein the lid and the base are configured to compress a grommet installed around the at least one cable when the lid is attached to the base.

In some embodiments, a system includes a roof deck; a passthrough installed on the roof deck, wherein the passthrough includes a base having a first end and a second end opposite the first end, a first surface extending from the first end to the second end, a second surface opposite the first surface, and an aperture extending from the first surface to the second surface, a lid attached to the base, and a sidewall between the base and the lid; a photovoltaic module installed on the roof deck, wherein the photovoltaic module includes a first side lap, wherein the first side lap includes an aperture; a roofing shingle installed on the roof deck, wherein the roofing shingle includes a first side lap, wherein the first side lap of the roofing shingle includes an aperture, wherein the first side lap of the roofing shingle overlays the first side lap of the photovoltaic module, wherein the aperture of the roofing shingle, the aperture of the photovoltaic module, and an aperture formed within the roof deck are aligned, and wherein the aperture of the base of the passthrough is aligned with the aperture of the roof deck, the aperture of the roofing shingle, and the aperture of the photovoltaic module; and at least one cable, wherein the at least one cable includes a first end and a second end opposite the first end of the at least one cable, and wherein the aperture of the base of the passthrough, the aperture of the roofing shingle, and the aperture of the photovoltaic module, are sized and shaped to receive the at least one cable, and wherein the first end of the at least one cable extends outwardly from the first end of base and the first end of the lid of the passthrough.

In some embodiments, the system includes a first flashing member, wherein the first flashing member is between the passthrough and the first side lap of the roofing shingle. In some embodiments, the system includes a second flashing member, wherein the second flashing member is between the first side lap of the roofing shingle and the first side lap of the photovoltaic module.

BRIEF DESCRIPTION OF THE DRAWINGS

FIGS. **1** and **2** illustrate some embodiments of an electrical cable passthrough;

FIGS. **3**A and **3**B illustrate some embodiments of an electrical cable passthrough, a photovoltaic module, and a roofing shingle installed on a roof deck;

FIGS. **3**C and **3**D illustrate some embodiments of a roofing shingle and a photovoltaic module shown in FIG. **3**A and FIG. **4**;

FIGS. **4** through **4**D illustrate some embodiments of an electrical cable passthrough and components thereof, a photovoltaic module, and a roofing shingle installed on a roof deck; and

FIGS. **5** and **6** illustrate some embodiments of an electrical cable passthrough installed on a roof deck.

DETAILED DESCRIPTION

Referring to FIGS. **1** and **2**, in some embodiments, an electrical cable passthrough **10** (hereinafter "passthrough **10**") is configured to be installed on a roof deck **100** of a structure. In some embodiments, the passthrough **10** is a component of a photovoltaic system. In some embodiments, the passthrough **10** includes a base **12** and a lid **14**. In some embodiments, the lid **14** is configured to be attached to the base **12**. In some embodiments, the lid **14** is configured to be removably attached to the base **12**.

In some embodiments, the base **12** includes a first end **11** and a second end **13** opposite the first end **11**. In some embodiments, the base **12** incudes a base plate **16**. In some embodiments, the base plate **16** includes a first surface **18** and a second surface **20** opposite the first surface **18**. In some embodiments, the first surface **18** extends from the first end **11** to the second end **13**. In some embodiments, the second surface **20** extends from the first end **11** to the second end **13**. In some embodiments, the base plate **16** has a circular shape. In some embodiments, the base plate **16** is substantially circular in shape. In some embodiments, the base plate **16** has a square shape. In some embodiments, the base plate **16** has a rectangular shape. In some embodiments, the base plate **16** has a triangular shape. In some embodiments, the base plate **16** has an elliptical shape. In some embodiments, the base plate **16** has a polygonal shape. In some embodiments, the base plate **16** has at least one aperture **22** extending from the first surface **18** to the second surface **20**. In some embodiments, the at least one aperture **22** includes a plurality of apertures **22**. In some embodiments, each of the apertures **22** is sized and shaped to receive a fastener **24**. In some embodiments, the fastener **24** is a screw. In some embodiments, the fastener **24** is a bolt. In some embodiments, the fastener **24** is a nail. In some embodiments, the fasteners **24** are configured to mount and secure the passthrough **10** on the roof deck **100**. In some embodiments, the base plate **16** is configured to be positioned substantially flush with the roof deck **100**. In some embodiments, the base plate **16** is configured to be positioned flush with the roof deck **100**.

In some embodiments, an aperture **26** extends from the first surface **18** to the second surface **20**. In some embodiments, the aperture **26** is circular in shape. In some embodiments, the aperture **26** is square in shape. In some embodiments, the aperture **26** is rectangular in shape. In some embodiments, the aperture **26** is polygonal in shape. In some embodiments, the base **12** includes a tubular portion **28** extending from the second surface **20**. In some embodiments, the aperture **26** extends from the first surface **18** to the tubular portion **28**. In some embodiments, the tubular portion **28** includes internal threads **30**. In some embodiments, the tubular portion **28** is sized and shaped to receive a conduit **110**. In some embodiments, external threads **112** of the conduit **110** threadedly engage the internal threads **30** of the tubular portion **28**. In some embodiments, the tubular portion **28** includes external threads and the conduit **110** includes internal threads that threadedly engage with the external threads of the tubular portion. In some embodiments, the conduit **110** is attached to the tubular portion **28** by a fastener. In some embodiments, the conduit **110** is attached to the tubular portion **28** by an adhesive. In some embodiments, the conduit **110** is attached to the tubular portion **28** by welding. In some embodiments, a seal **111** is located below the base **12**. In some embodiments, the seal **111** includes a first portion **113**. In some embodiments, the first portion **113** is located below the base **12**. In some embodiments, the first portion **113** is sized and shaped to match or substantially match the size and shape of the lower surface **15** of the base **12**. In some embodiments, the seal **111** includes a second portion **115**. In some embodiments, the second portion **115** extends from the first portion **113**. In some embodiments, the second portion **115** is located below the sidewall **32**. In some embodiments, the second portion **115** is sized and shaped to match or substantially match the size and shape of the lower surface of the sidewall **32**. In some embodiments, the seal **111** is composed of butyl, silicone, rubber, epoxy, latex, neoprene, or polyurethane foam.

In some embodiments, the base **12** includes a sidewall **32**. In some embodiments, the sidewall **32** extends outwardly from the base plate **16**. In some embodiments, the sidewall **32** extends outwardly from the first surface **18** of the base plate **16**. In some embodiments, the sidewall **32** includes a first portion **34**. In some embodiments, the first portion **34** circumferentially surrounds the aperture **26** at the first surface **18**. In some embodiments, the first portion **34** is substantially circular in shape. In some embodiments, the sidewall **32** includes a second portion **36**. In some embodiments, the second portion **36** is adjacent the first portion **34**. In some embodiments, the second portion **36** extends from the first portion **34** to the first end **11**. In some embodiments, the second portion **36** is oblong in shape. In some embodiments, the second portion **36** is substantially rectangular in shape. In some embodiments, the first portion **34** and the second portion **36** form a keyhole shape.

In some embodiments, the second portion **36** includes a bore **35**. In some embodiments, the bore **35** includes an internal thread. In some embodiments, the bore **35** is configured to receive a fastener **37**. As discussed below, in some embodiments, the fastener **37** is configured to fasten the lid **14** to the base **12**. In some embodiment, the second portion **36** includes at least one tab **39** extending outwardly proximate the second end **13**. In some embodiments, the at least one tab **39** includes a plurality of tabs **39**. As discussed below, in some embodiments, the at least one tab **39** is sized and shaped to engage a slot **58** of the lid **14**.

In some embodiments, the sidewall **32** includes a first end **38**. In some embodiments, the first end **38** of the sidewall **32** is proximate to the first end **11** of the base **12**. In some embodiments, at least one first cavity **40** is located at the first end **38** of the sidewall **32**. In some embodiments, the at least one first cavity **40** includes one first cavity **40**. In some embodiments, the at least one first cavity **40** includes a plurality of first cavities **40**. In some embodiments, the plurality of first cavities **40** includes two of the first cavities **40**. In some embodiments, the plurality of first cavities **40** includes three of the first cavities **40**. In some embodiments, the plurality of first cavities **40** includes four of the first cavities **40**. In some embodiments, the plurality of first cavities **40** includes more than four of the cavities **40**. In some embodiments, each of the first cavities **40** is semi-tubular or semi-cylindrical in shape.

In some embodiments, the lid **14** includes a first end **42** and a second end **44** opposite the first end **42**. In some embodiments, the lid **14** includes an outer wall **46**. In some embodiments, the outer wall **46** includes a first portion **48**. In some embodiments, the first portion **48** is substantially circular in shape. In some embodiments, the outer wall **46** includes a second portion **50**. In some embodiments, the second portion **50** is adjacent the first portion **48**. In some embodiments, the second portion **50** extends from the first portion **48** to the first end **42**. In some embodiments, the second portion **50** is oblong in shape. In some embodiments, the second portion **50** is substantially rectangular in shape. In some embodiments, the first portion **48** and the second portion **50** form a keyhole shape. In some embodiments, the lid **14** is attached to the sidewall **32** of the base **12**. In some embodiments, the lid **14** has a size and shape that is substantially similar to the size and shape of the sidewall **32** of the base **12**. In some embodiments, the first portion **48** of the lid **14** covers the first portion **34** of the sidewall **32** of the base **12**. In some embodiments, the second portion **50** of the lid **14** covers the second portion **36** of the sidewall **32** of the base **12**. In some embodiments, the first end **42** of the lid **14** is substantially aligned with the first end **11** of the base **12**. In some embodiments, the first end **42** of the lid **14** is aligned with the first end **11** of the base **12**. In some embodiments, the lid **14** is hollow and with the base **12** forms an interior portion **51**. In some embodiments, the interior portion **51** is sized and shaped to receive at least one electrical cable **120**. In some embodiments, the passthrough includes a O-ring **57**. In some embodiments, the O-ring **57** is between the base **12** and the lid **14**. In some embodiments, the O-ring **57** is between the sidewall **32** of the base **12** and the outer wall **46** of the lid. In some embodiments, the O-ring **57** forms a watertight seal between the base **12** and the lid **14**. In some embodiments, the O-ring **57** is made from rubber. In some embodiments, the O-ring **57** is made from styrene-butadiene or styrene-butadiene rubber (SBR). In some embodiments, the O-ring **57** is made from nitrile butadiene rubber (NBR). In some embodiments, the O-ring **57** is made from EPDM rubber (ethylene propylene diene monomer rubber). In some embodiments, the O-ring **57** is made from plastic. In some embodiments, the O-ring **57** is made from a polymer. In some embodiments, the O-ring **57** is made from silicone. In some embodiments, the O-ring **57** is made from polytetrafluoroethylene (PTFE). In some embodiments, the O-ring **57** is made from fluorosilicone (FVMQ). In some embodiments, the O-ring **57** is made from polyurethane.

In some embodiments, the lid **14** includes an inner wall **53** and a receptacle **55** extending from the inner wall **53**. In some embodiments, the receptacle **55** includes an internal thread. In some embodiments, the bore **35** and the receptacle **55** are configured to receive the fastener **37** to fasten the lid **14** to the base **12**. In some embodiments, the fastener **37** is a screw. In some embodiments, the fastener **37** is a compression screw. In some embodiments, the fastener **37** is bolt. In some embodiments, the fastener **37** is a threaded rod.

In some embodiments, at least one second cavity **52** is located at the first end **42** of the lid **14**. In some embodiments, the at least one second cavity **52** includes a single cavity **52**. In some embodiments, the at least one second cavity **52** includes a plurality of second cavities **52**. In some embodiments, the plurality of second cavities **52** includes two of the second cavities **52**. In some embodiments, the plurality of second cavities **52** includes three of the second cavities **52**. In some embodiments, the plurality of second cavities **52** includes four of the second cavities **52**. In some embodiments, the plurality of second cavities **52** includes more than four of the second cavities **52**. In some embodiments, each of the second cavities **52** is semi-tubular or semi-cylindrical in shape.

In some embodiments, at least one aperture **54** is located at the first end **38** of the sidewall **32**. In some embodiments, the at least one aperture **54** of the sidewall **32** is sized and shaped to receive a corresponding one of the at least one electrical cable **120**. In some embodiments, the at least one first cavity **40** of the base **12** and the at least one second cavity **52** of the lid **14** are sized and shaped to form a corresponding one of the at least one aperture **54**. In some embodiments, the at least one aperture **54** includes a plurality of apertures **54**. In some embodiments, the plurality of first cavities **40** of the base **12** and the plurality of second cavities **52** of the lid **14** are sized and shaped to form a corresponding one of the a plurality of the apertures **54**. In some embodiments, each of the plurality of the apertures **54** is sized and shaped to receive a corresponding one of the a plurality of the electrical cables **120**. In some embodiments, the plurality of the apertures **54** includes two of the apertures **54**. In some embodiments, the plurality of the electrical cables **120** includes two of the electrical cables **120**. In some embodiments, the plurality of the apertures **54** includes three of the apertures **54**. In some embodiments, the plurality of the electrical cables **120** includes three of the electrical cables **120**. In some embodiments, the plurality of the apertures **54** includes four of the apertures **54**. In some embodiments, the plurality of the electrical cables **120** includes four of the electrical cables **120**. In some embodiments, the plurality of the apertures **54** includes more than four of the apertures **54**. In some embodiments, the plurality of the electrical cables **120** includes more than four of the electrical cables **120**. In some embodiments, the at least one aperture **54** is circular in shape. In some embodiments, the at least one aperture **54** is square in shape.

In some embodiments, the lid **14** includes at least one slot **58** at the second end **44** thereof that is sized and shaped to receive a corresponding one of the at least one tab **39** of the base **12**. In some embodiments, the at least one slot **58** includes a plurality of slots **58**. In some embodiments, the at least one tab **39** includes a plurality of tabs **39**. In some embodiments, each of the tabs **39** is sized and shaped to engage a corresponding one of the slots **58** to removably secure the lid **14** to the base **12**.

In some embodiments, one or both of the base **12** and the lid **14** are made from plastic. In some embodiments, the base **12** and the lid **14** are made from polypropylene. In some embodiments, the base **12** and the lid **14** are made from a polymer blend with polypropylene as a base resin. In another embodiment, the base **12** and the lid **14** are made from thermoplastic polyolefin (TPO). In another embodiment, the TPO is a modified TPO including fiberglass and/or other filler material. In another embodiment, the base **12** and the lid **14** are made from metal. In some embodiments, the base **12** and the lid **14** are made of aluminum. In another embodiment, the base **12** and the lid **14** are made of stainless steel.

In some embodiments, the at least one electrical cable **120** includes the at least one electrical connector **122**. In some embodiments, the at least one electrical connector **122** is a bulkhead (i.e., panel mount) connector. In some embodiments, the at least one electrical connector **122** is electrically connected to jumper modules, jumper cables, junction boxes, or module leads.

In some embodiments, a grommet **60** is installed on the at least one electrical cable **120**. In some embodiments, a grommet **60** is installed proximate to a first end **121** of the at least one electrical cable **120**. In some embodiments, the grommet **60** includes at least one opening **62**. In some embodiments, the at least one opening **62** is sized and shaped to receive a corresponding one of the at least one electrical cable **120**. In some embodiments, the grommet **60** is installed and located between the at least one first cavity **40** of the base **12** and the at least one second cavity **52** of the lid **14**. In some embodiments, the grommet **60** is configured to be compressed between the at least one first cavity **40** of the base **12** and the at least one second cavity **52** of the lid **14** when the lid **14** is attached to the base **12**. In some embodiments, when the fastener **37** is tightened to attached the lid **14** to the base **12**, the grommet **60** is compressed. In some embodiments, a user may selectively tighten or loosen the fastener **37** to control the amount of force on and compression of the grommet **60**. In some embodiments, the compression of the grommet **60** forms a watertight seal with the electrical cables **120** to inhibit the entry of water or moisture in the interior portion **51** of the passthrough **10**. In some embodiments, the grommet **60** is elastically deformed. In some embodiments, the grommet **60** is made from rubber. In some embodiments, the grommet **60** is made from styrene-butadiene or styrene-butadiene rubber (SBR). In some embodiments, the grommet **60** is made from nitrile butadiene rubber (NBR). In some embodiments, the grommet **60** is made from EPDM rubber (ethylene propylene diene monomer rubber). In some embodiments, the grommet **60** is made from plastic. In some embodiments, the grommet **60** is made from a polymer. In some embodiments, the grommet **60** is made from silicone. In some embodiments, the grommet **60** is made from polytetrafluoroethylene (PTFE). In some embodiments, the grommet **60** is made from fluorosilicone (FVMQ). In some embodiments, the grommet **60** is made from polyurethane.

In some embodiments, the at least one opening **62** of the grommet **60** includes a plurality of openings **62**. In each of the plurality of openings **62** is sized and shaped to receive a corresponding one of the plurality of the electrical cables **120**. In some embodiments, the grommet **60** includes a plurality of sections **64**. In some embodiments, each of the plurality of sections **64** has a tubular shape. In some embodiments, the plurality of sections **64** are integral. In some embodiments, the grommet **60** has a figure eight shape. In some embodiments, each of the plurality of openings **62** is located in a corresponding one of the plurality of sections **64**. In some embodiments, each of the plurality of sections **64** of the grommet **60** is located between a corresponding one of the plurality of first cavities **40** of the base **12** and a corresponding one of the plurality of second cavities **52** of the lid **14**. In some embodiments, the plurality of openings **62** is arranged in a linear arrangement. In some embodiments, the grommet **60** has a figure eight shape.

In some embodiments, the passthrough **10** has a thickness of 1 inch to 10 inches. In some embodiments, the passthrough **10** has a thickness of 1 inch to 9 inches. In some embodiments, the passthrough **10** has a thickness of 1 inch to 8 inches. In some embodiments, the passthrough **10** has a thickness of 1 inch to 7 inches. In some embodiments, the passthrough **10** has a thickness of 1 inch to 6 inches. In some embodiments, the passthrough **10** has a thickness of 1 inch to 5 inches. In some embodiments, the passthrough **10** has a thickness of 1 inch to 4 inches. In some embodiments, the passthrough **10** has a thickness of 1 inch to 3 inches. In some embodiments, the passthrough **10** has a thickness of 1 inch to 2 inches. In some embodiments, the passthrough **10** has a thickness of 2 inches to 10 inches. In some embodiments, the passthrough **10** has a thickness of 2 inches to 9 inches. In some embodiments, the passthrough **10** has a thickness of 2 inches to 8 inches. In some embodiments, the passthrough **10** has a thickness of 2 inches to 7 inches. In some embodiments, the passthrough **10** has a thickness of 2 inches to 6 inches. In some embodiments, the passthrough **10** has a thickness of 2 inches to 5 inches. In some embodiments, the passthrough **10** has a thickness of 2 inches to 4 inches. In some embodiments, the passthrough **10** has a thickness of 2 inches to 3 inches.

In some embodiments, the passthrough **10** has a thickness of 3 inches to 10 inches. In some embodiments, the passthrough **10** has a thickness of 3 inches to 9 inches. In some embodiments, the passthrough **10** has a thickness of 3 inches to 8 inches. In some embodiments, the passthrough **10** has a thickness of 3 inches to 7 inches. In some embodiments, the passthrough **10** has a thickness of 3 inches to 6 inches. In some embodiments, the passthrough **10** has a thickness of 3 inches to 5 inches. In some embodiments, the passthrough **10** has a thickness of 3 inches to 4 inches. In some embodiments, the passthrough **10** has a thickness of 4 inches to 10 inches. In some embodiments, the passthrough **10** has a thickness of 4 inches to 9 inches. In some embodiments, the passthrough **10** has a thickness of 4 inches to 8 inches. In some embodiments, the passthrough **10** has a thickness of 4 inches to 7 inches. In some embodiments, the passthrough **10** has a thickness of 4 inches to 6 inches. In some embodiments, the passthrough **10** has a thickness of 4 inches to 5 inches.

In some embodiments, the passthrough **10** has a thickness of 5 inches to 10 inches. In some embodiments, the passthrough **10** has a thickness of 5 inches to 9 inches. In some embodiments, the passthrough **10** has a thickness of 5 inches to 8 inches. In some embodiments, the passthrough **10** has a thickness of 5 inches to 7 inches. In some embodiments, the passthrough **10** has a thickness of 5 inches to 6 inches. In some embodiments, the passthrough **10** has a thickness of 6 inches to 10 inches. In some embodiments, the passthrough **10** has a thickness of 6 inches to 9 inches. In some embodiments, the passthrough **10** has a thickness of 6 inches to 8 inches. In some embodiments, the passthrough **10** has a thickness of 6 inches to 7 inches. In some embodiments, the passthrough **10** has a thickness of 7 inches to 10 inches. In some embodiments, the passthrough **10** has a thickness of 7 inches to 9 inches. In some embodiments, the passthrough **10** has a thickness of 7 inches to 8 inches. In some embodiments, the passthrough **10** has a thickness of 8 inches to 10 inches. In some embodiments, the passthrough **10** has a thickness of 8 inches to 9 inches. In some embodiments, the passthrough **10** has a thickness of 9 inches to 10 inches.

In some embodiments, the passthrough **10** has a thickness of 1 inch. In some embodiments, the passthrough **10** has a thickness of 2 inches. In some embodiments, the passthrough **10** has a thickness of 3 inches. In some embodiments, the passthrough **10** has a thickness of 4 inches. In some embodiments, the passthrough **10** has a thickness of 5 inches. In some embodiments, the passthrough **10** has a thickness of 6 inches. In some embodiments, the passthrough **10** has a thickness of 7 inches. In some embodiments, the passthrough **10** has a thickness of 8 inches. In some embodiments, the passthrough **10** has a thickness of 9 inches. In some embodiments, the passthrough **10** has a thickness of 10 inches.

Referring to FIGS. **3**A and **3**B, in some embodiments, the passthrough **10** is installed on the roof deck **100**. In some embodiments, the roof deck **100** includes an aperture **101**. Referring to FIGS. **3**A through **3**D, in some embodiments, at least one roofing shingle **200***a* (also referred to as a roofing module) is installed on the roof deck **100**. In some embodiments, the roofing shingle **200***a* does not include a solar cell, a photovoltaic cell, or any electrical device that converts the energy of light into electricity. In some embodiments, the roofing shingle **200***a* does not include any electrically active components. In some embodiments, the roofing shingle **200***a* is a cuttable roofing shingle. In some embodiments, the roofing shingle **200***a* is a nailable roofing shingle. In some embodiments, the roofing shingle **200***a* is attached to the roof deck **100** by at least one fastener or a plurality of fasteners. In some embodiments, the fasteners include nails, rivets, screws or staples, or a combination thereof. In some embodiments, the roofing shingle **200***a* is textured. In some embodiments, the roofing shingle **200***a* includes a printed pattern **203**. In some embodiments, the printed pattern **203** includes a depiction of solar cells. In some embodiments, the at least one roofing shingle **200***a* includes a plurality of roofing shingles **200***a*. In some embodiments, each of the roofing shingles **200***a* includes a first end **202***a* and a second end **204***a* opposite the first end **202***a*. In some embodiments, each of the roofing shingles **200***a* includes a first side lap **206***a* at the first end **202***a*. In some embodiments, each of the roofing shingles **200***a* includes a second side lap **208***a* at the second end **204***a*. In some embodiments, the first side lap **206***a* of the roofing shingle **200***a* includes an aperture **210***a* extending from an upper surface to a lower surface thereof. In some embodiments, the roofing shingle **200***a* includes a head lap portion **215***a*. In some embodiments, the head lap portion **215***a* is configured to receive one or more of the fasteners for installing the roofing shingle **200***a* to the roof deck. In some embodiments, at least a portion of a first roofing shingle **200***a* is configured to overlay the head lap portion **215***a* of a second roofing shingle **200***a*.

In some embodiments, the roofing shingle **200***a* includes a structure, composition and/or function of similar to those of more or one of the embodiments of the roofing modules disclosed in in U.S. application Ser. No. 17/831,307, filed Jun. 2, 2022, titled "Roofing Module System," and published under U.S. Patent Application Publication No. 2022-0393637 on Dec. 8, 2022, and owned by GAF Energy LLC; and/or U.S. application Ser. No. 18/169,718, filed Feb. 15, 2023, titled "Roofing Module System," and published under U.S. Patent Application Publication No. 2023-0203815 on Jun. 29, 2023, and owned by GAF Energy LLC, the disclosure and contents of each of which are incorporated by reference herein in their entirety.

In some embodiments, at least one photovoltaic module **200***b* is installed on the roof deck **100**. In some embodiments, the photovoltaic module **200***b* includes a plurality of solar cells **201**. In some embodiments, the at least one photovoltaic module **200***b* includes a plurality of photovoltaic modules **200***b*. In some embodiments, each of the photovoltaic modules **200***b* includes a first end **202***b* and a second end **204***b* opposite the first end **202***b*. In some embodiments, each of the photovoltaic modules **200***b* includes a first side lap **206***b* at the first end **202***b*. In some embodiments, each of the photovoltaic modules **200***b* includes a second side lap **208***b* at the second end **204***b*. In some embodiments, the second side lap **208***b* of the photovoltaic module **200***b* includes an aperture **212***b* extending from an upper surface to a lower surface thereof.

In some embodiments, the photovoltaic module **200***b* includes a head lap portion **215***b*. In some embodiments, the head lap portion **215***b* is configured to receive one or more fasteners. In some embodiments, at least a portion of a first photovoltaic module **200***b* is configured to overlay the head lap portion **215***b* of a second photovoltaic module **200***b*.

In some embodiments, at least a portion of a first photovoltaic module **200***b* is configured to overlay the head lap portion **215***a* of a first roofing shingle **200***a*. In some embodiments, at least a portion of a first roofing shingle **200***a* is configured to overlay the head lap portion **215***b* of a first photovoltaic module **200***b*.

In some embodiments, each of the photovoltaic modules **200***b* includes a structure, composition, features, components, and/or function similar to those of one or more embodiments of the photovoltaic modules disclosed in PCT International Patent Publication No. WO 2022/051593, Application No. PCT/US2021/049017, published Mar. 10, 2022, entitled Building Integrated Photovoltaic System, and owned by GAF Energy LLC; and U.S. Pat. No. 11,251,744 to Bunea et al., issued Feb. 15, 2022, entitled "Photovoltaic Shingles and Methods of Installing Same," and owned by GAF Energy, LLC, the disclosure and contents of each of which are incorporated by reference herein in their entirety.

In some embodiments, the first side lap **206***a* of the roofing shingle **200***a* overlays the second side lap **208***b* of the photovoltaic module **200***b*. In some embodiments, the aperture **101** of the roof deck **100**, the aperture **210***a* of the roofing shingle **200***a*, and the aperture **212***b* of the photovoltaic module **200***b* are aligned with one another. In some embodiments, the aperture **101** of the roof deck **100**, the aperture **210***a* of the roofing shingle **200***a*, and the aperture **212***b* of the photovoltaic module **200***b* are substantially aligned with one another.

In some embodiments, the passthrough **10** is installed on the first side lap **206***a* of the roofing shingle **200***a*. In some embodiments, the base **12** of the passthrough **10** is substantially parallel to the roof deck **100**. In some embodiments, a lower surface **15** of the base **12** is substantially parallel to the roof deck **100**. In some embodiments, the aperture **26** of the base **12** of the passthrough **10** is configured to align with the aperture **101** of the roof deck **100**, the aperture **210***a* of the roofing shingle **200***a*, and the aperture **212***b* of the photovoltaic module **200***b*.

Referring to FIGS. **3**A and **3**B, in some embodiments, the first side lap **206***a* of the roofing shingle **200***a* and the second side lap **208***b* of the photovoltaic module **200***b* form a wireway **220**. In some embodiments, the passthrough **10** is located within the wireway **220** when the passthrough **10** is installed on the roof deck **100**. In some embodiments, a cover **222** is attached to the wireway **220**. In some embodiments, the cover **222** is removably attached to the wireway **220**. In some embodiments, the cover **222** covers the passthrough **10**. In some embodiments, the cover **222** inhibits water or moisture from entering the wireway **220**. In some embodiments, the cover **222** includes multiple cover portions or a plurality of the covers **222** connected to one another. In some embodiments, each of the wireway **220** and the cover **222** has a structure, composition, components, features, and/or function similar to those of one or more embodiments of the wireways and covers disclosed in PCT International Patent Publication No. WO 2022/051593, Application No. PCT/US2021/049017, published Mar. 10, 2022, entitled Building Integrated Photovoltaic System, owned by GAF Energy LLC; and/or U.S. Pat. No. 11,251,744 to Bunea et al., issued Feb. 15, 2022, entitled "Photovoltaic Shingles and Methods of Installing Same," and owned by GAF Energy, LLC, the disclosure and contents of each of which are incorporated by reference herein in their entirety.

In some embodiments, an underlayment layer overlays the roof deck **100**. In some embodiments, the electrical cable passthrough **10** is installed on the underlayment layer. In some embodiments, a sealant is applied intermediate the underlayment layer and the base **12** of the electrical cable passthrough **10**. In some embodiments, the sealant includes butyl, silicone, rubber, epoxy, latex, neoprene, or polyurethane foam.

In some embodiments, the interior portion **51** is sized and shaped to receive the electrical cables **120**. In some embodiments, the electrical cables **120** are connected to corresponding ones of the electrical connectors **122**. In some embodiments, the electrical cables **120** extend though the passthrough **10** and the conduit **110**. In some embodiments, the aperture **26** of the base **12** of the passthrough **10** is sized and shaped to receive the electrical cables **120**. In some embodiments, the first ends **121** of the electrical cables **120** extend outwardly from the first end **11** of base **12** and the first end **42** of the lid **14** of the passthrough **10**. In some embodiments, each of the electrical cables **120** is a THHN stranded wire. In some embodiments, each of the electrical cables **120** is a XHHW stranded wire.

Still referring to FIGS. **3**A and **3**B, in some embodiments, a receiver **150** includes a first end **152**, a second end **154** opposite the first end **152**, and an aperture **156** extending from the first end **152** to the second end **154**. In some embodiments, the receiver **150** has an symmetric shape. In some embodiments, the receiver **150** is tubular in shape. In some embodiments, the receiver **150** has a rectangular-tubular shape. In some embodiments, the receiver **150** has an asymmetric shape. In some embodiments, the receiver **150** has a curvilinear shape. In some embodiments, the receiver **150** has an elbow-shape.

In some embodiments, the receiver **150** is installed underneath an inner surface **102** of the roof deck **100**. In some embodiments, the first end **152** of the receiver **150** is attached to the tubular portion **28** of the base **12** of the passthrough **10**. In some embodiments, the first end **152** of the receiver **150** is attached a fitting **158** which, in turn, is attached to the tubular portion **28** of the base **12** of the passthrough **10**. In some embodiments, the second end **154** of the receiver **150** is attached to an end of the conduit **110**. In some embodiments, the aperture **156** of the receiver **150** is sized and shaped to receive the at least one electrical cable **120** therethrough. In some embodiments, a rafter **104** is located underneath the inner surface **102** roof deck **100**. In some embodiments, the rafter **104** supports the roof deck **100**. In some embodiments, the rafter **104** is perpendicular to the roof deck **100**. In some embodiments, the rafter **104** is substantially perpendicular to the roof deck **100**. In some embodiments, the receiver **150** is installed proximate to a rafter **104**. In some embodiments, the receiver **150** is installed proximate to a first surface **106** of the rafter **104**. In some embodiments, the receiver **150** is installed adjacent to a first surface **106** of the rafter **104**. In some embodiments, the receiver **150** is configured to carry the at least one electrical cable **120** away from the rafter **104**. In some embodiments, the receiver **150** is configured to clamp the passthrough **10** to the roof deck **100**.

In some embodiments, the receiver **150** extends 30 degrees to 90 degrees relative to the inner surface **102** of the roof deck **100**. In another embodiment, the receiver **150** extends 30 degrees to 80 degrees relative to the inner surface **102** of the roof deck **100**. In another embodiment, the receiver **150** extends 30 degrees to 70 degrees relative to the inner surface **102** of the roof deck **100**. In another embodiment, the receiver **150** extends 30 degrees to 60 degrees relative to the inner surface **102** of the roof deck **100**. In another embodiment, the receiver **150** extends 30 degrees to 50 degrees relative to the inner surface **102** of the roof deck **100**. In another embodiment, the receiver **150** extends 30 degrees to 40 degrees relative to the inner surface **102** of the roof deck **100**.

In some embodiments, the receiver **150** extends 40 degrees to 90 degrees relative to the inner surface **102** of the roof deck **100**. In another embodiment, the receiver **150** extends 40 degrees to 80 degrees relative to the inner surface **102** of the roof deck **100**. In another embodiment, the receiver **150** extends 40 degrees to 70 degrees relative to the inner surface **102** of the roof deck **100**. In another embodiment, the receiver **150** extends 40 degrees to 60 degrees relative to the inner surface **102** of the roof deck **100**. In another embodiment, the receiver **150** extends 40 degrees to 50 degrees relative to the inner surface **102** of the roof deck **100**. In some embodiments, the receiver **150** extends 50 degrees to 90 degrees relative to the inner surface **102** of the roof deck **100**. In another embodiment, the receiver **150** extends 50 degrees to 80 degrees relative to the inner surface **102** of the roof deck **100**. In another embodiment, the receiver **150** extends 50 degrees to 70 degrees relative to the inner surface **102** of the roof deck **100**. In another embodiment, the receiver **150** extends 50 degrees to 60 degrees relative to the inner surface **102** of the roof deck **100**.

In some embodiments, the receiver **150** extends 60 degrees to 90 degrees relative to the inner surface **102** of the roof deck **100**. In another embodiment, the receiver **150** extends 60 degrees to 80 degrees relative to the inner surface **102** of the roof deck **100**. In another embodiment, the receiver **150** extends 60 degrees to 70 degrees relative to the inner surface **102** of the roof deck **100**. In some embodiments, the receiver **150** extends 70 degrees to 90 degrees relative to the inner surface **102** of the roof deck **100**. In another embodiment, the receiver **150** extends 70 degrees to 80 degrees relative to the inner surface **102** of the roof deck **100**. In some embodiments, the receiver **150** extends 80 degrees to 90 degrees relative to the inner surface **102** of the roof deck **100**.

In some embodiments, a second one of the roofing shingle **200***a* may be located in place of the photovoltaic module **200***b*. In some embodiments, the roofing shingle **200***a* may overlap at least a portion of another shingle **200***a* that is vertically adjacent and below same. In some embodiments, at least a portion of the roofing shingle **200***a* may be overlapped by yet another roofing shingle **200***a* that is vertically adjacent and above same. In some embodiments, the roofing shingle **200***a* may overlap at least a portion of another photovoltaic module **200***b* that is vertically adjacent and below same. In some embodiments, at least a portion of the roofing shingle **200***a* may be overlapped by another photovoltaic module **200***b* that is vertically adjacent and above same.

FIGS. **4** through **4**D show some embodiments of a passthrough **310** and components thereof installed on the roof deck **100**. In some embodiments, the passthrough **310** and the overall roofing system include a structure, function, components and features similar to those of the passthrough **10** and the corresponding system, with some differences. Referring to FIGS. **4**A and **4**B, in some embodiments, the passthrough **310** includes a base plate **311**, a sidewall **332** and a tubular member **328**.

Referring to FIG. **4**, in some embodiments, the roofing system includes a first flashing member **312**. In some embodiments, the roofing system includes a second flashing member **314**. Referring to FIGS. **4**, **4**C and **4**D, in some embodiments, the first flashing member **312** includes a base **316**, an aperture **318** through the base **316**, and a sidewall **320** around at least a portion of a perimeter of the base **316**. In some embodiments, the aperture **318** is pre-formed in the base **318** during manufacturing thereof. In some embodiments, the base **316** may have one or more templates for a user to cut out material therefrom the form the aperture **318** for custom installation or alignment. In some embodiments, the first flashing member **312** includes a tab **315** extend from one end thereof.

In some embodiment, the first flashing member **312** overlays and is above the first side lap **206***a* of the roofing shingle **200***a*. In some embodiments, a separate seal or sealing member is between the underside of the first flashing member **312** and the first side lap **206***a* of the roofing shingle **200***a*. In some embodiments, the seal is similar to the seal **111**. In some embodiments, the aperture **318** is aligned or substantially aligned with the aperture **210***a* of the roofing shingle **200***a*. In some embodiments, the passthrough **310** overlays and is above the first flashing member **312**. In some embodiments, the tubular member **328** of the passthrough **310** is installed within the aperture **318** of the first flashing member **312**. In some embodiments, the first flashing member **312** is configured to watershed water. In some embodiments, the sidewall **320** channels a flow of any water down the slope of the roof deck **100**. In some embodiments, the sidewall **320** diverts water away from the seal member. In some embodiments, another roofing shingle overlays the tab **315** of the first flashing member **312** to facilitate securement of the first flashing member **312** on the roof deck.

In some embodiments, the second flashing member **314** is located between the roofing shingle **200***a* and the photovoltaic module **200***b*. In some embodiments, the second flashing member **314** is located between the lower surface of the first side lap **206***a* of the roofing shingle **200***a* and the upper surface of the photovoltaic module **200***b*. In some embodiments, the second flashing member **314** is located between the lower surface of the first side lap **206***a* of the first roofing shingle **200***a* and the upper surface of the second side lap **208***b* of the photovoltaic module **200***b*. In some embodiments, the second flashing member **314** includes an aperture **322**. In some embodiments, the aperture **322** is aligned or substantially aligned with the aperture **318** of the first flashing member **312**, the aperture **210***a* of the roofing shingle **200***a*, the aperture **212***b* of the photovoltaic module **200***b*, and the aperture **101** of the roof deck **100**. In some embodiments, the second flashing member **314** includes an upper edge **325**. In some embodiments, the upper edge **325** is angled relative to side portions thereof. In some embodiments, the second flashing member **314** is configured to seal to prevent water and/or watershed water. In some embodiments, a cover **360** covers the passthrough **310** and the first flashing member **312** to watershed water.

In some embodiments, the tubular member **328** of the passthrough **310** is installed within and through the aperture **322** of the second flashing member **314**, the aperture **318** of the first flashing member **312**, the aperture **210***a* of the roofing shingle **200***a*, the aperture **212***b* of the photovoltaic module **200***b*, and the aperture **101** of the roof deck **100**. In some embodiments, a conduit **350** extends through the foregoing apertures and connects to the tubular member **328** of the passthrough **310** and houses and shrouds the electrical wires. In some embodiments, the upper edge **325** of the second flashing member **314** is configured to prevent any water that enters between the second side lap **208***b* of the photovoltaic module **200***b* and the underside of the second flashing member **314** from travelling laterally (left to right) and diverts and drains the water from right to left.

Referring to FIGS. **5** and **6**, in some embodiments, an electrical box **400** is installed underneath the roof deck **100** and within the structure. In some embodiment, the electrical box **400** is installed on the inner surface **102** of the roof deck **100**. In some embodiment, the electrical box **400** is installed proximate to the first surface **106** of the rafter **104**. In some embodiments, the electrical box **400** directly receives the tubular portion **28** of the passthrough **10**. In some embodiments, the conduit **110** is not utilized. In some embodiments, the electrical box **400** is a combiner box.

In some embodiments, the passthrough **10** may include a structure, composition, function, and/or one or more features similar to those of more or one of the embodiments of the electrical cable passthrough devices disclosed in U.S. Pat. No. 11,177,639 to Nguyen et al., issued Nov. 16, 2021, entitled Electrical Cable Passthrough for Photovoltaic Systems, owned by GAF Energy LLC; and/or U.S. Pat. No. 11,658,470, to Nguyen et al. issued May 23, 2023, entitled Electrical Cable Passthrough for Photovoltaic Systems, owned by GAF Energy LLC, the disclosure and contents of each of which are incorporated by reference herein in their entirety.

In some embodiments, the passthrough **10** is configured for use with electrical systems of building-integrated photovoltaic (BIPV) systems. In another embodiment, the passthrough **10** is configured for use with retrofit photovoltaic systems for roofing. In other embodiments, the passthrough **10** is configured for use with other electrical systems.

The embodiments described herein are merely exemplary and that a person skilled in the art may make many variations and modifications without departing from the spirit and scope of the invention. All such variations and modifications are intended to be included within the scope of the invention.

What is claimed is:

1. A system, comprising:

a roof deck;

a passthrough installed on the roof deck, wherein the passthrough includes

a base having

a first end and a second end opposite the first end,
a first surface extending from the first end to the second end,
a second surface opposite the first surface,
an aperture extending from the first surface to the second surface, and
a plurality of first cavities,
a lid attached to the base,
wherein the lid includes
a first end substantially aligned with the first end of the base, and
a plurality of second cavities,
a sidewall between the base and the lid,
wherein the sidewall includes
a first end, wherein the first end of the sidewall is proximate to the first end of the base, and
a plurality of apertures,
wherein the plurality of apertures of the sidewall is located at the first end of the sidewall,
wherein each of the plurality of first cavities and each of a corresponding one of the plurality of second cavities are sized and shaped to form a corresponding one of the plurality of apertures,
wherein the first surface is juxtaposed with the roof deck, and
wherein the aperture of the base of the passthrough is configured to align with an aperture formed within the roof deck;
a plurality of cables,
wherein each of the plurality of cables includes a first end and a second end opposite the first end of the cable,
wherein each of the plurality of apertures of the sidewall is sized and shaped to receive a corresponding one of the plurality of cables; and
a grommet installed on the at least one plurality of cables proximate to the first end of each of the plurality of cables,
wherein the grommet is located between the plurality of first cavities and the plurality of second cavities,
wherein the grommet includes
a plurality of openings sized and shaped to receive a corresponding one of the plurality of cables, and
a plurality of sections, wherein each of the plurality of openings is located in a corresponding one of the plurality of sections,
wherein each of the plurality of sections of the grommet is located between a corresponding one of the plurality of first cavities and a corresponding one of the plurality of second cavities,
wherein the lid and the base are configured to compress the grommet between the plurality of first cavities and the plurality of second cavities and around the plurality of cables when the lid is attached to the base,
wherein the aperture of the base of the passthrough is sized and shaped to receive the plurality of cables, and
wherein the first end of each of the plurality of cables extends outwardly from the first end of base and the first end of the lid of the passthrough.

2. The system of claim **1**, wherein the lid is removably attached to the base.

3. The system of claim **1**, further comprising a photovoltaic module installed on the roof deck, wherein the passthrough is installed proximate to the photovoltaic module.

4. The system of claim **1**, wherein each of the plurality of cables includes an electrical connector, wherein each of the plurality of apertures of the sidewall is sized and shaped to receive a corresponding one of the electrical connector of the corresponding one of the plurality of cables.

5. The system of claim **1**, wherein the base of the passthrough includes a tubular member extending from the first surface, wherein the tubular member includes the aperture of the base, and wherein the aperture of the roof deck is sized and shaped to receive the tubular member.

6. The system of claim **5**, further comprising a receiver attached to the tubular member.

7. The system of claim **5**, further comprising an electrical box installed underneath the roof deck, and wherein the electrical box directly receives the tubular member.

8. A cable passthrough, comprising:
a base having
a first end and a second end opposite the first end,
a first surface extending from the first end to the second end,
a second surface opposite the first surface,
a plurality of first cavities, and
an aperture extending from the first surface to the second surface;
a lid attached to the base,
wherein the lid includes
a first end substantially aligned with the first end of the base, and
a plurality of second cavities; and
a sidewall between the base and the lid,
wherein the sidewall includes
a first end, wherein the first end of the sidewall is proximate to the first end of the base, and
a plurality of apertures,
wherein the plurality of apertures of the sidewall is located at the first end of the sidewall,
wherein each of the plurality of first cavities and a corresponding one of each of the plurality of second cavities are sized and shaped to form a corresponding one of the plurality of apertures,
wherein the cable passthrough is configured to be installed on a roof deck such that the first surface is juxtaposed with the roof deck, and
wherein the aperture of the base is configured to align with an aperture formed within the roof deck,
wherein the cable passthrough is configured to receive a plurality of cables,
wherein the aperture of the base is sized and shaped to receive the plurality of cables,
wherein the plurality of cables is capable of extending outwardly from the first end of the base and the first end of the lid of the cable passthrough, and
wherein the lid and the base are configured to compress a grommet installed between the plurality of first cavities and the plurality of second cavities and around the plurality of cables when the lid is attached to the base,
wherein the grommet includes a plurality of openings sized and shaped to receive a corresponding one of the plurality of cables, and a plurality of sections, wherein each of the plurality of openings is located in a corresponding one of the plurality of sections, wherein each of the plurality of sections is configured to be located between a corresponding one of the plurality of first cavities and a corresponding one of the plurality of second cavities.

9. A system, comprising:
a passthrough installed on a roof deck, wherein the passthrough includes
a base having a plurality of first cavities,
a lid attached to the base, wherein the lid includes a plurality of second cavities, and
a sidewall between the base and the lid, wherein the sidewall includes a plurality of apertures, wherein each of the plurality of first cavities and a corresponding one of each of the plurality of second cavities are sized and shaped to form a corresponding one of the plurality of apertures;
a plurality of cables,
wherein each of the plurality of apertures of the sidewall receives a corresponding one of the plurality of cables; and
a grommet installed on the plurality of cables, wherein the grommet is located between the plurality of first cavities and the plurality of second cavities, wherein the grommet includes
a plurality of sections, and
a plurality of openings, each of which is located in a corresponding one of the plurality of sections,
wherein each of the plurality of openings receives a corresponding one of the plurality of cables,
wherein each of the plurality of sections of the grommet is located between a corresponding one of the plurality of first cavities and a corresponding one of the plurality of second cavities, and
wherein the lid and the base are configured to compress the grommet between the plurality of first cavities and the plurality of second cavities and around the plurality of cables when the lid is attached to the base.

10. The system of claim **9**, wherein the lid is removably attached to the base.

11. The system of claim **9**, further comprising a photovoltaic module installed on the roof deck, wherein the passthrough is installed proximate to the photovoltaic module.

12. The system of claim **9**, wherein each of the plurality of cables includes an electrical connector, wherein each of the plurality of apertures of the sidewall is sized and shaped to receive a corresponding one of the electrical connector of the plurality of cables.

13. The system of claim **9**, wherein the base of the passthrough includes a tubular member, and wherein an aperture in the roof deck is sized and shaped to receive the tubular member.

14. The system of claim **13**, further comprising a receiver attached to the tubular member.

15. The system of claim **13**, further comprising an electrical box installed underneath the roof deck, and wherein the electrical box directly receives the tubular member.

* * * * *